Lecture 1:

Data vs. information: Concepts & ontology

Application of Computational Linguistics
2026-2027

Concepts

Concepts → mental representations

- help us understand the world around us
- are the building blocks of thoughts

- **Abstraction**:

  Concepts generalize from specific examples. For instance, the concept of "dog" includes all breeds, sizes, and colors.

- **Support human reasoning**:
  - We group things together:

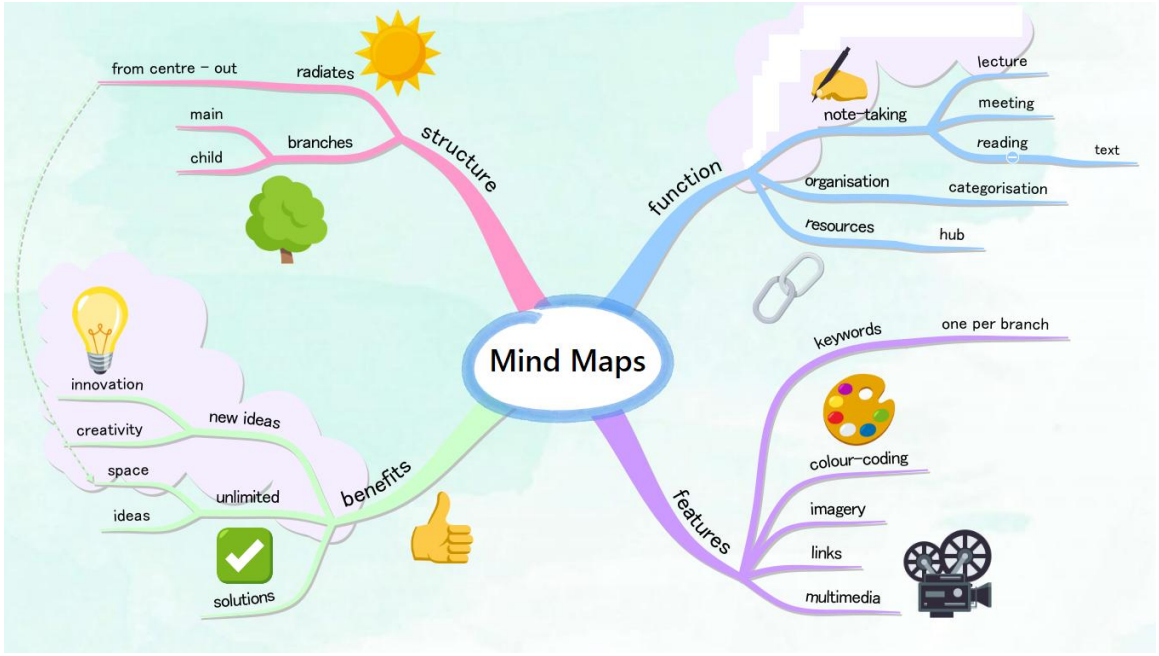
    elephants, tigers, lions are *animals* , apples are not
  - We draw inferences:

    If you know what an *animal* is, you can guess that a ***wombat*** is an animal even if it is the first time you see one.

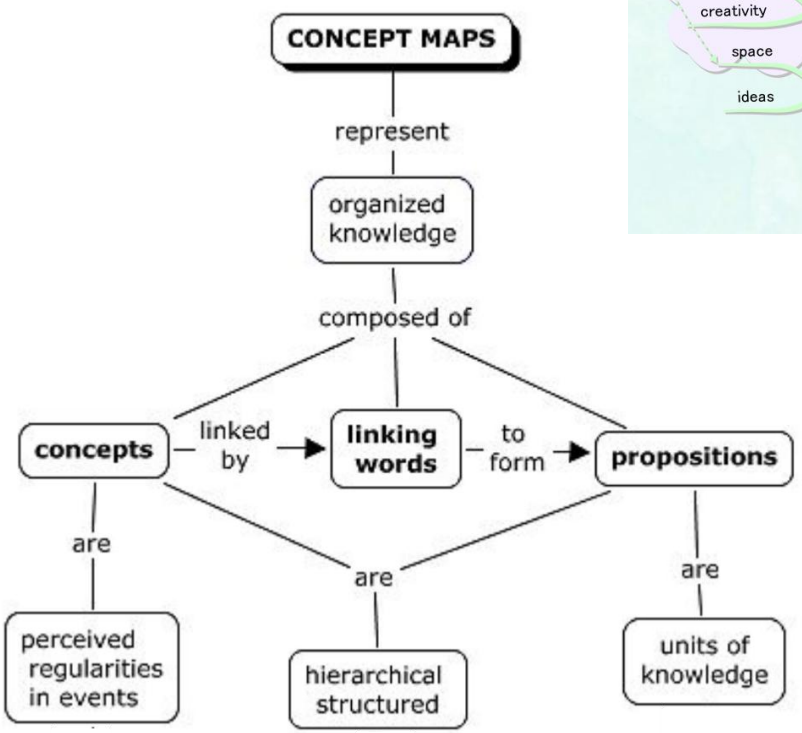


Charles J. Sharp,
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via Wikimedia Commons

Concepts



What ideas relate to this topic?



How are the concepts related?

Cognitive maps

Cognitive maps are the umbrella term for all kind of visual representations of mental models, and encompass both concept maps and mind maps.

Adapted after:
https://www.ayoa.com/imindmap/blog/mind-maps-vs-concept-maps/
Orue, Amalia & Álvarez, Gonzalo & Vitini, Fausto. (2008). Using Concept Maps to improve Scientific Communications.

Mind map or Concept map ?

Single central topic	vs	Multiple concepts
Tree-like		Network-like
Idea generation		Knowledge representation
Low complexity		Medium to High complexity
Surface to moderate understanding depth		Deep understanding

Use cases:

Lecture notes
Brainstorming sessions
Personal learning
Knowledge management
Ontology design
Project planning

Concepts in AI

Concepts are the glue between raw data and intelligent behaviour.

In Machine Learning (Statistical Models)

- Concepts are **learned from data** rather than manually defined.
- Concepts are represented as **points in high-dimensional space**.
- Example: A spam filter learns the concept of “spam” by analysing thousands of labelled emails.

In Symbolic AI (Logic-Based Systems)

- Concepts are **explicitly defined** using rules, logic, and ontologies.
- Example: In a medical expert system, the concept of "fever" might be defined as "body temperature > 38°C".

In Ontologies and Knowledge Graphs concepts are structured into **networks of meaning**.

Symbolic AI

.

this presentation uses slides by:

- Valentina Tamma, A. Gomez Perez, N. Noy, D. McGuinness, E. Kendal, A. Rector and O. Corcho
- Olivier Dameron, Alan Rector, Nick Drummond, Matthew Horridge
- Olivier Bodenreider

A definition of an ontology (1998):

So what is an ontology then?

An ontology is a (formal), explicit specification of a shared conceptualisation

formal: an ontology should be machine-readable

shared: an ontology captures consensual knowledge, that is not private to some individual, but accepted by a group

explicit: the types of concepts used, and the constraints on their use are explicitly defined

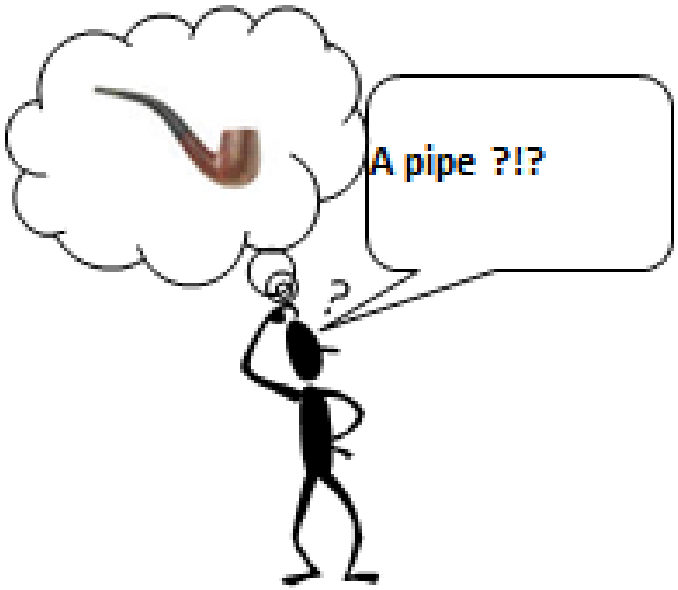
conceptualisation: an abstract model of some phenomenon in the world which identifies the relevant concepts of that phenomenon

T. Gruber, 1993; R. Studer, V. R. Benjamins, and D. Fensel, 1998

Ontological commitment

Agreement on the meaning of the vocabulary used to share knowledge.





Unified Medical Language System (UMLS)

- a unified resource for machine interpretation of **medical language**.
- developed by the **U.S. National Library of Medicine (NLM)**
- introduced in the 1960s

Medical text contains:

- Synonyms
- Abbreviations
- Spelling variations
- Multiple naming

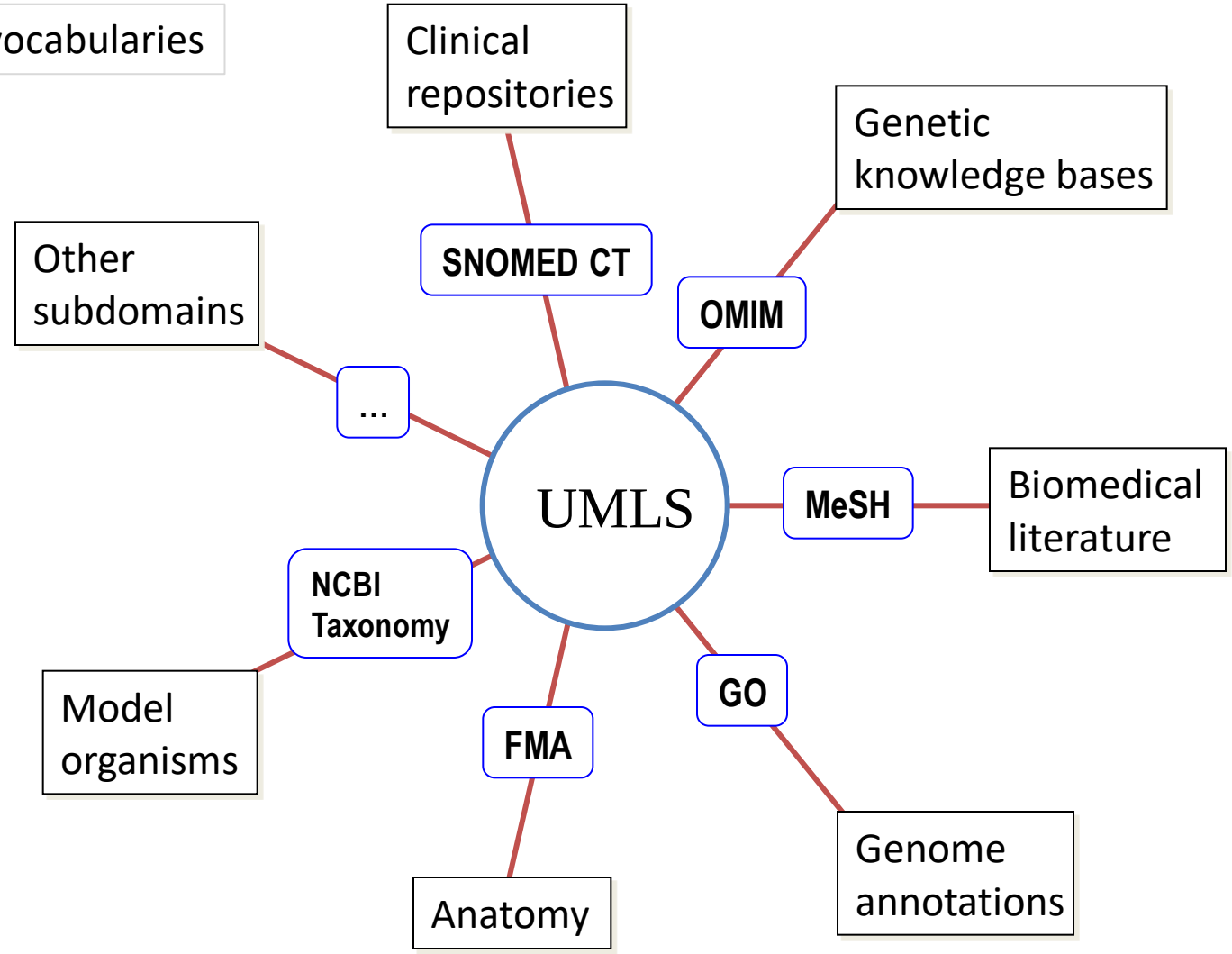
Example:

- Heart Attack
- Myocardial Infarction
- MI

but same concept

Unified Medical Language System

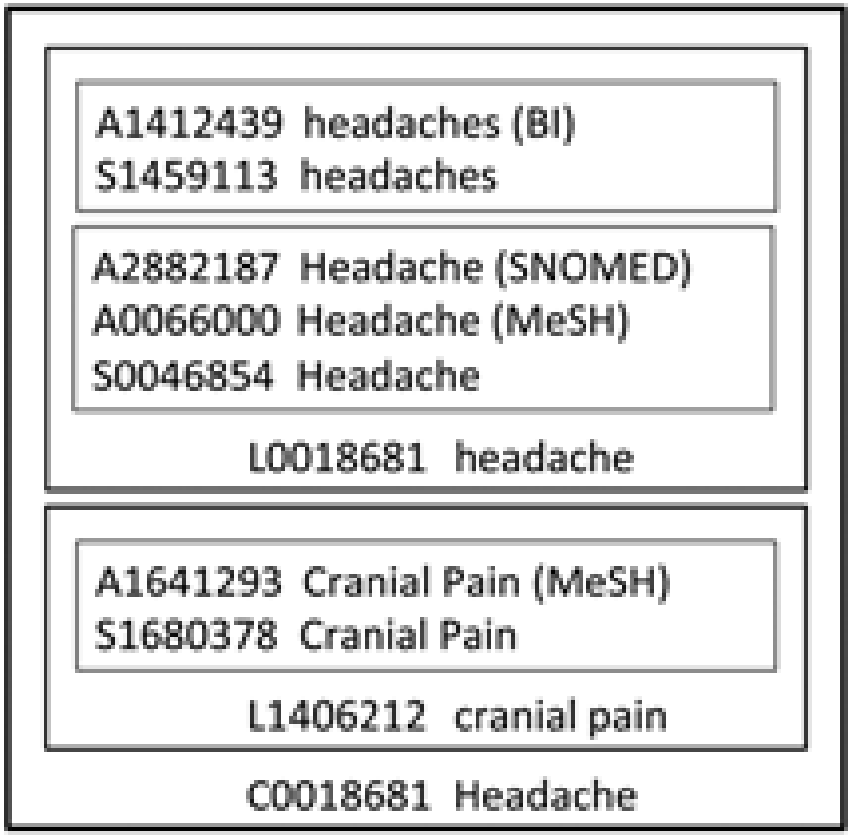
168 source vocabularies



Unified Medical Language System (UMLS)

- SUI : String Unique Identifier
- AUI : Atom UI
- LUI : Lexical (term) UI
- CUI : concept UI

- e.g.



UMLS: 3 components

- Lexicon
  - lexical items
  - Part of speech and variant information

Eg: aspirin
- Base form: *aspirin*
- Part of speech: noun
- Inflectional variants: *aspirins* (plural)
- Derivational variants: *aspirinic* (adjective form)
- Spelling variants: --

- Meta-thesaurus
  - Integrates biomedical vocabularies.
  - Each concept links to its source vocabularies, allowing cross-referencing

Eg.: aspirin
- CUI: C0004057
- Preferred name: Aspirin
- Synonyms: Acetylsalicylic Acid, ASA, ...
- Source vocabularies: MeSH, SNOMED CT,...
- Semantic Type: Pharmacologic Substance, Organic Chemical

- Semantic Network
  - high-level categories (sematic types)
  - relations among them

Eg.: aspirin
Semantic Relations: Treats, Has Ingredient (via type-level)
- Semantic Type: Pharmacologic Substance, Organic Chemical
- Pharmacologic Substance treats Disease or Syndrome
- Organic Chemical has ingredient Clinical Drug

Each concept in the Meta-thesaurus is assigned one or more semantic types.
e.g. **fever** is of semantic type
- 'Disease or Syndrome',
- 'Sign or Symptom' ,
- 'Finding'

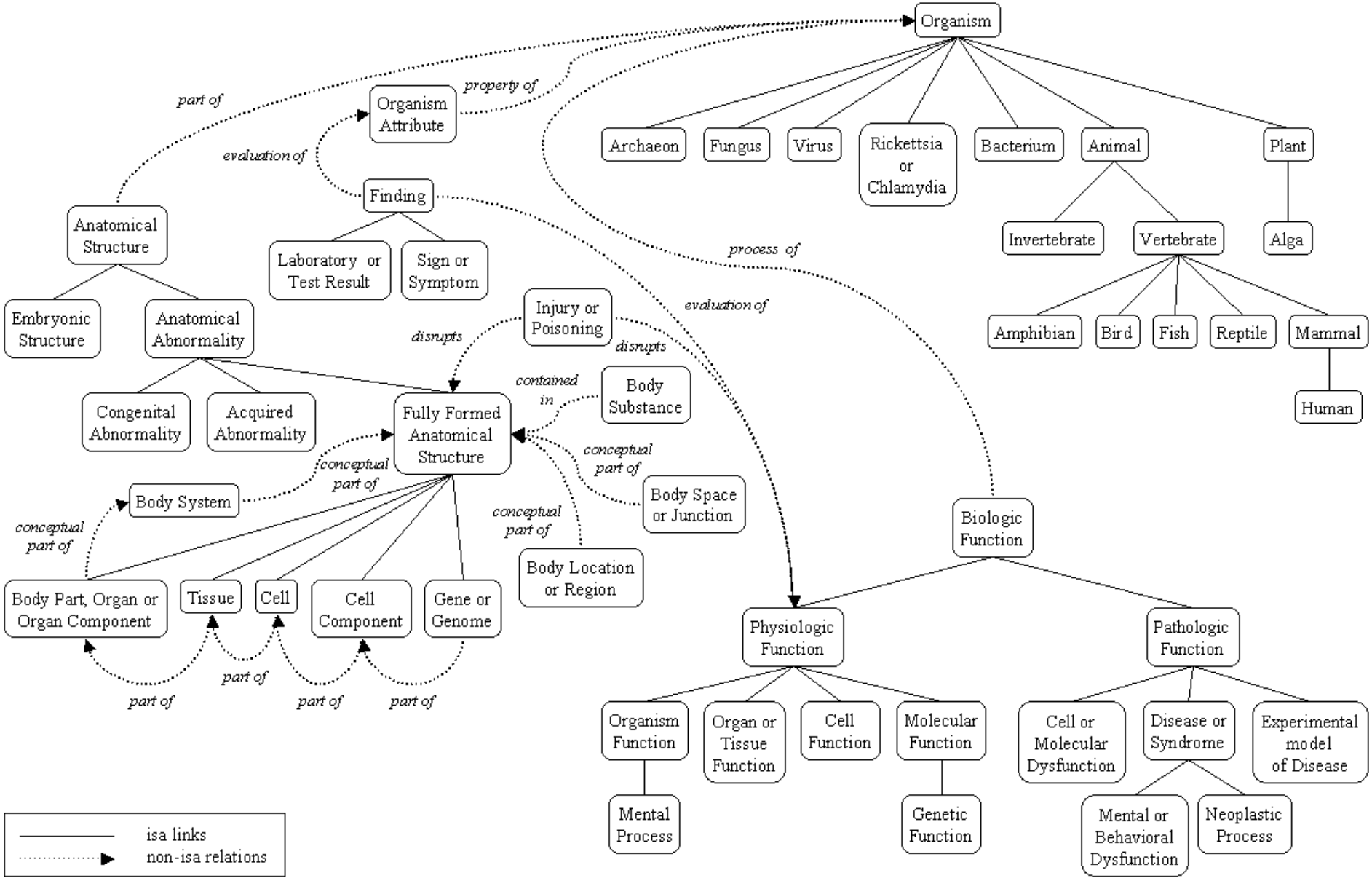
http://umls.it.ilstu.edu/umls.html

UMLS: 3 components

- Lexicon
  - over 100,000 lexical entries
  - including both single-word and multi-word terms
- Meta-thesaurus
  - more than 60 families of biomedical terminologies
  - ~ 2.5 million concepts or terms
  - ~ 12 million relations among these concepts
- Semantic Network
  - 135 high-level categories (sematic types)
  - 54 relations among them

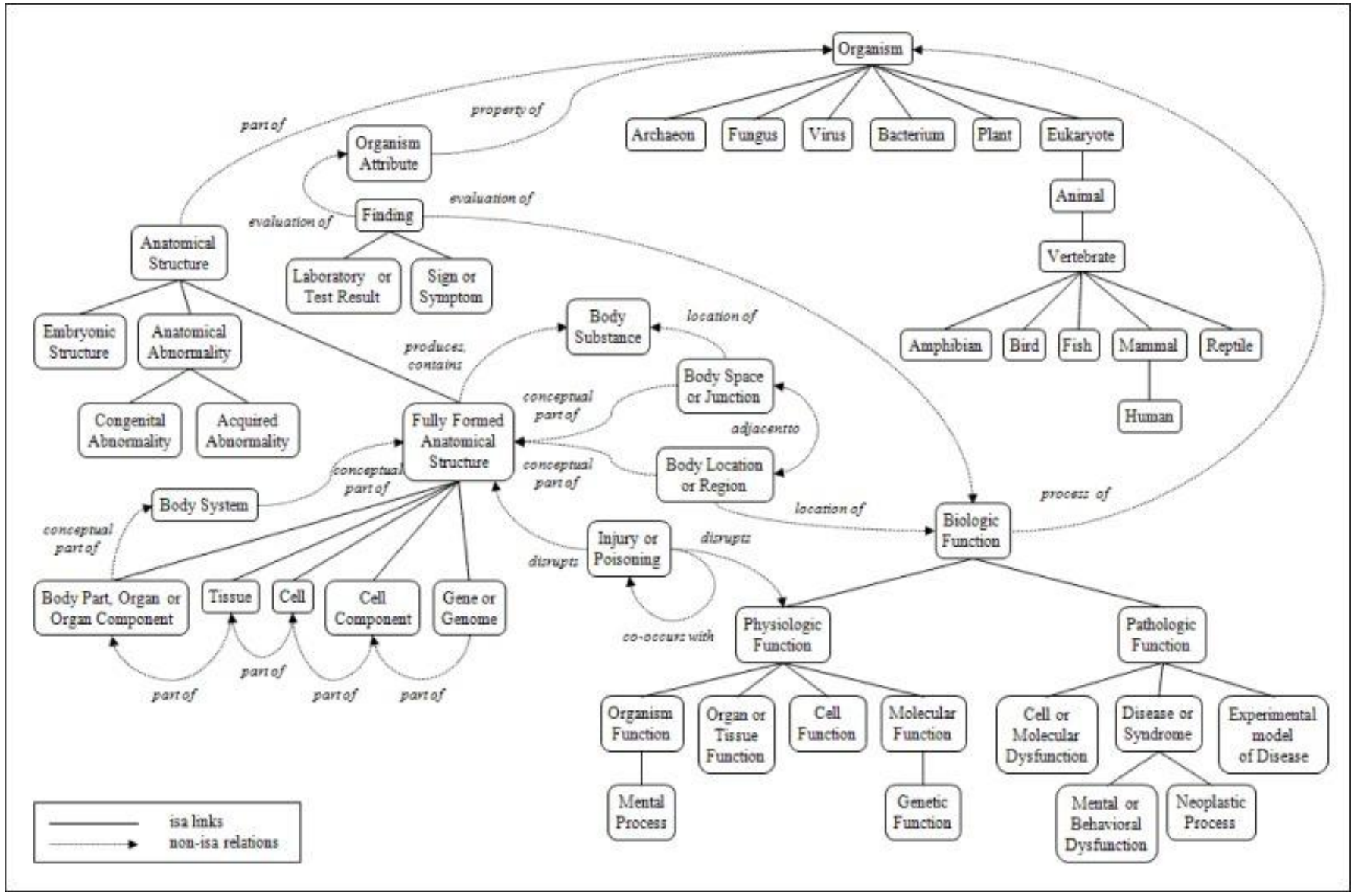
http://umls.it.ilstu.edu/umls.html

A subset of UMLS sematic network



http://umls.it.ilstu.edu/umls.html *accessed: 2020*

A subset of UMLS sematic network



https://www.ncbi.nlm.nih.gov/books/NBK9679/ *accessed: 2026*

UMLS Semantic Network Browser

The UMLS Semantic Network is one of three UMLS Knowledge Sources developed as part of the Unified Medical Language System project. The network provides a consistent categorization of all concepts represented in the UMLS Metathesaurus. More information...

List | Tree

- (A) Entity
  - (A1) Physical Object
    - (A1.1) Organism
      - (A1.1.1) Archaeon
      - (A1.1.2) Bacterium
      - + (A1.1.3) Eukaryote
      - (A1.1.4) Virus
    - (A1.2) Anatomical Structure
      - (A1.2.1) Embryonic Structure
      - + (A1.2.2) Anatomical Abnormality
      - + (A1.2.3) Fully Formed Anatomical Structure
    - + (A1.3) Manufactured Object
    - + (A1.4) Substance
  - + (A2) Conceptual Entity
- + (B) Event
- (H) isa
- + (R) associated_with

Anatomical Structure

Definition

A normal or pathological part of the anatomy or structural organization of an organism.

Properties

Unique Identifier: T017

Tree Number: A1.2

Usage Note: Few concepts will be assigned to this broad type.

Children

Embryonic Structure
Fully Formed Anatomical Structure
Anatomical Abnormality

Relations

Anatomical Structure isa Physical Object
Anatomical Structure location_of Bacterium
Anatomical Structure location_of Fungus
Anatomical Structure location_of Virus
Anatomical Structure part_of Organism

Inverse Relations

Anatomical Abnormality isa Anatomical Structure
Embryonic Structure isa Anatomical Structure
Fully Formed Anatomical Structure isa Anatomical Structure

Inherited Relations

https://uts.nlm.nih.gov/uts/umls/semantic-network/T017

MeSH - Medical Subject Headings (e.g.1)

source: https://meshb.nlm.nih.gov/record/ui?ui=D006261

Headache MeSH Descriptor Data 2018

Details | Qualifiers | MeSH Tree Structures | Concepts

MeSH Heading	Headache
Tree Number(s)	C23.888.592.612.441
Unique ID	D006261
Annotation	HEADACHE DISORDERS and specifics are also available but do not diagnose: use word of text
Scope Note	The symptom of PAIN in the cranial region. It may be an isolated benign occurrence or manifestation of a wide variety of HEADACHE DISORDERS.
Entry Term(s)	Bilateral Headache Cephalalgia Cephalgia Cephalodynia Cranial Pain Generalized Headache Head Pain Hemicrania Ocular Headache Orthostatic Headache Periorbital Headache Retro-Ocular Headache Sharp Headache Throbbing Headache Unilateral Headache Vertex Headache
NLM Classification #	WL 342
See Also	Headache Disorders
Public MeSH Note	2006 (1963); for HEMICRANIA see MIGRAINE 1963-2005
History Note	2006 (1963); for HEMICRANIA use MIGRAINE 1963-2005
Date Established	2006/01/01
Date of Entry	1999/01/01
Revision Date	2017/06/20

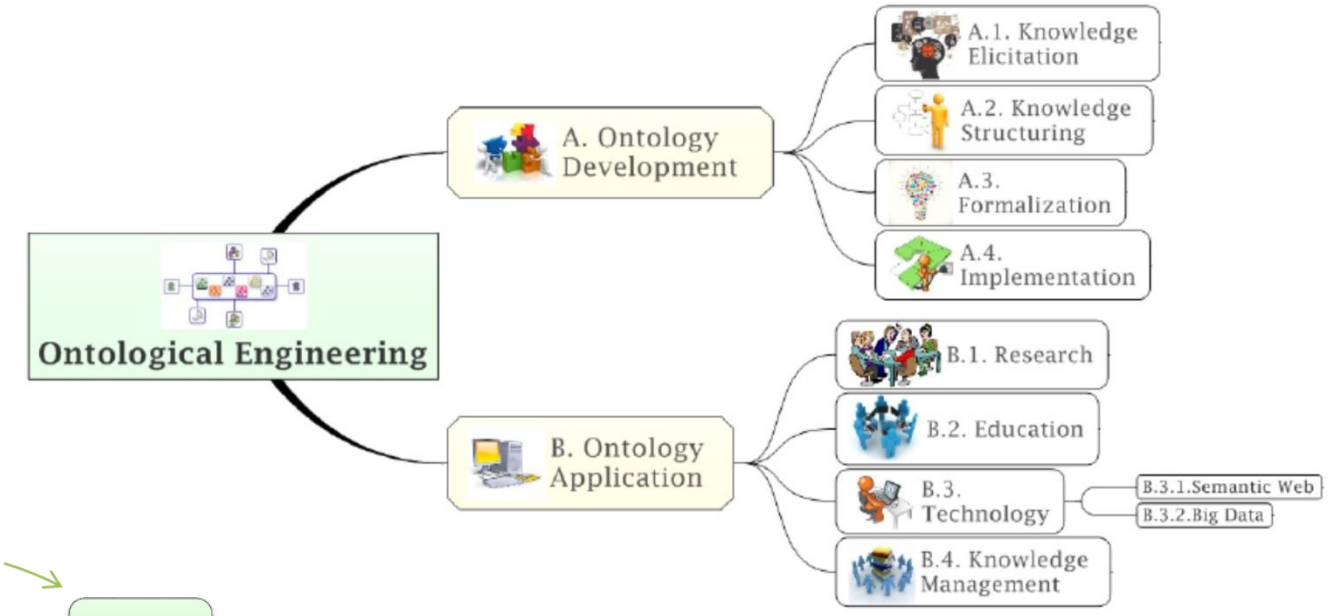
- MeSH (e.g.2)

source: https://meshb.nlm.nih.gov/record/ui?ui=D00626

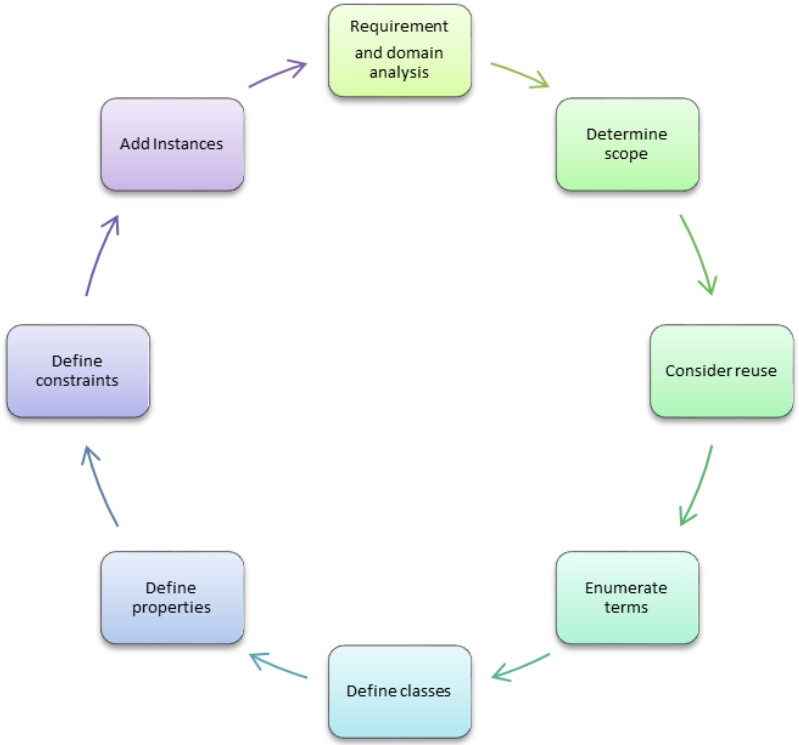
See also:
https://meshb.nlm.nih.gov/treeView



Ontology design



Big Data Structuring: The Role of Visual Models and Ontologies - Scientific Figure on ResearchGate. Available from: https://www.researchgate.net/figure/Ontological-Engineering_fig1_262691744 [accessed 1 Oct, 2020]



Source: Valentina Tamma

IDE and ontologies

- some people prefer simple text editors
- others prefer an IDE
- Protege is a popular IDE

  http://protege.stanford.edu

OWL ⇒ OWL DL semantics

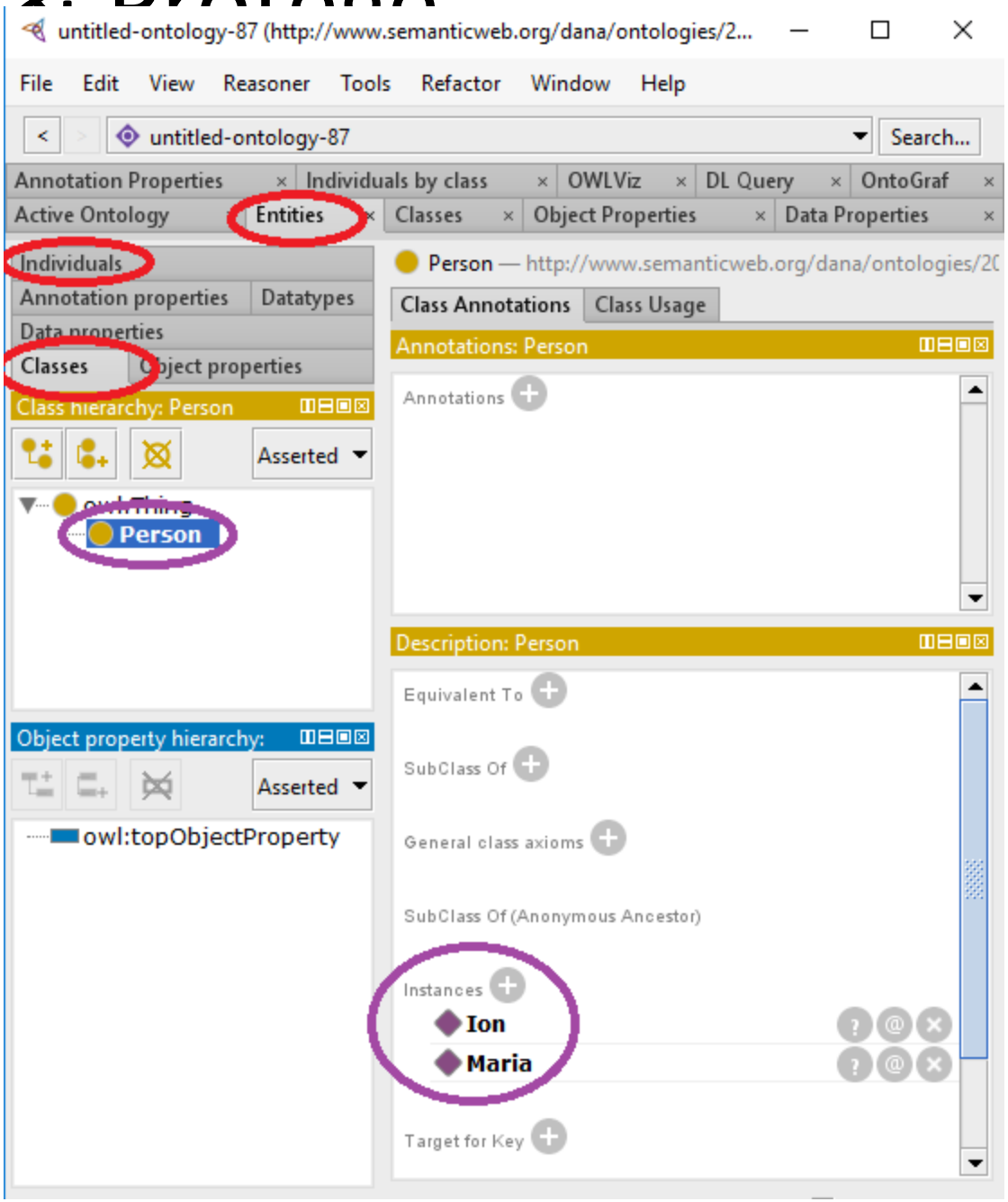
- OWL: Web Ontology Language

  - a family of knowledge representation languages

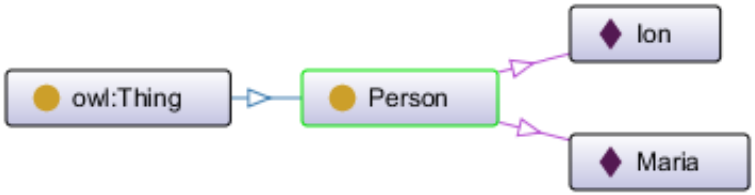
- DL: Description logics

OWL Elements & Protégé

- individuals
- classes



OWL family of languages supports a variety of syntaxes



RDF/XML

…

prefix …

Functional

- Declaration(Class(:Person))
- Declaration(
  NamedIndividual(:Ion))
- Declaration(
  NamedIndividual(:Maria))

…

Individual: :Ion (:Ion)

ClassAssertion(:Person :Ion)

Individual: :Maria (:Maria)

ClassAssertion(:Person :Maria)

Manchester

- Class: Person

- Individual: Ion
  Types: Person
- Individual: Maria
  Types: Person

… *and others* …

OWL Elements

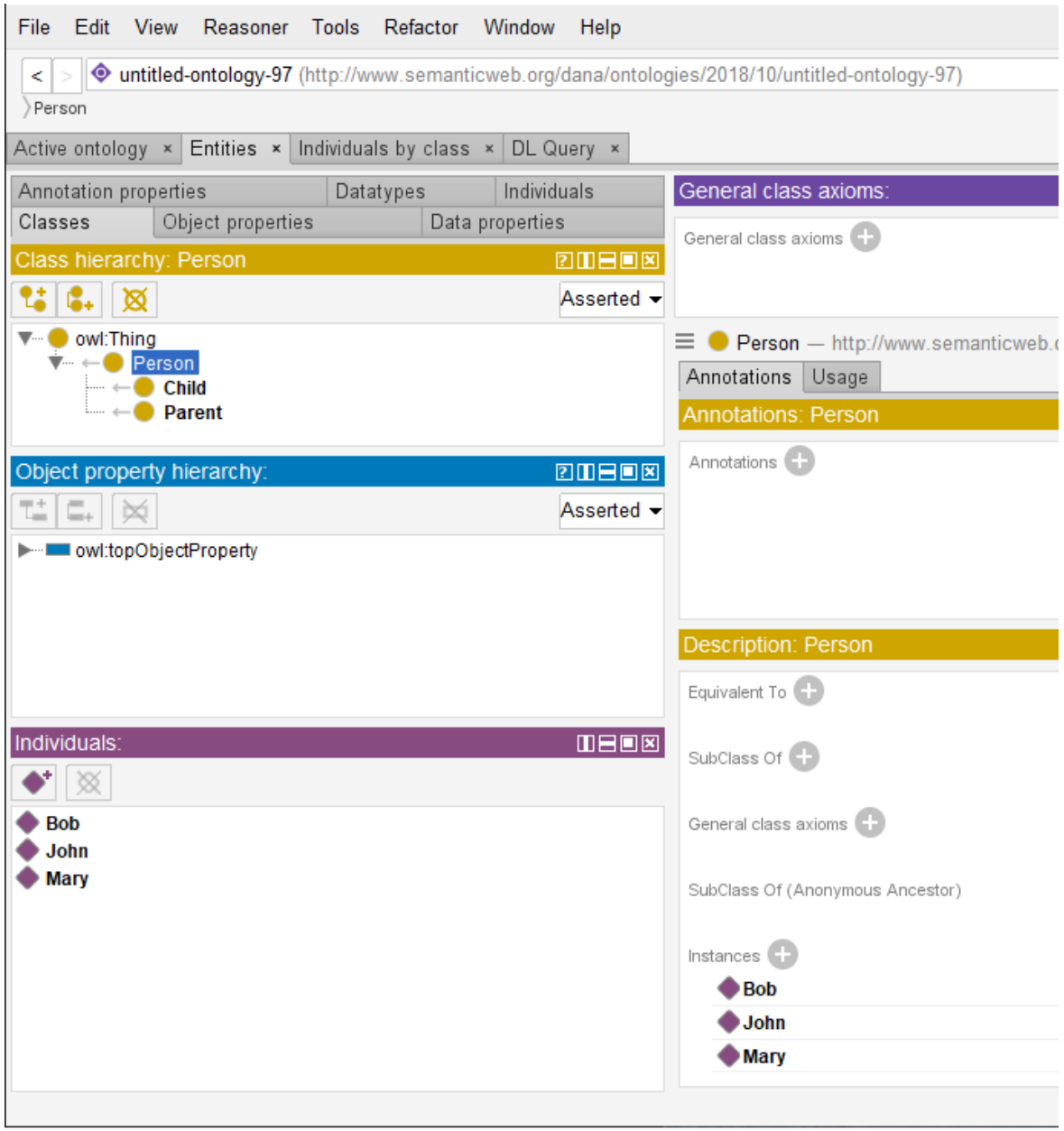
- individuals
- classes
  - special classes
  - operations
    - union, intersection, complement
  - properties (as sets)
    - subset, disjointness
- Properties (binary relations)
  - Domain, Range
  - sub-properties
  - Characteristics
    - Transitivity
    - Symmetry
    - Functional (0 or 1 – inRange)

Defining classes:
extensional vs. intensional approaches



OWL Elements & Protégé

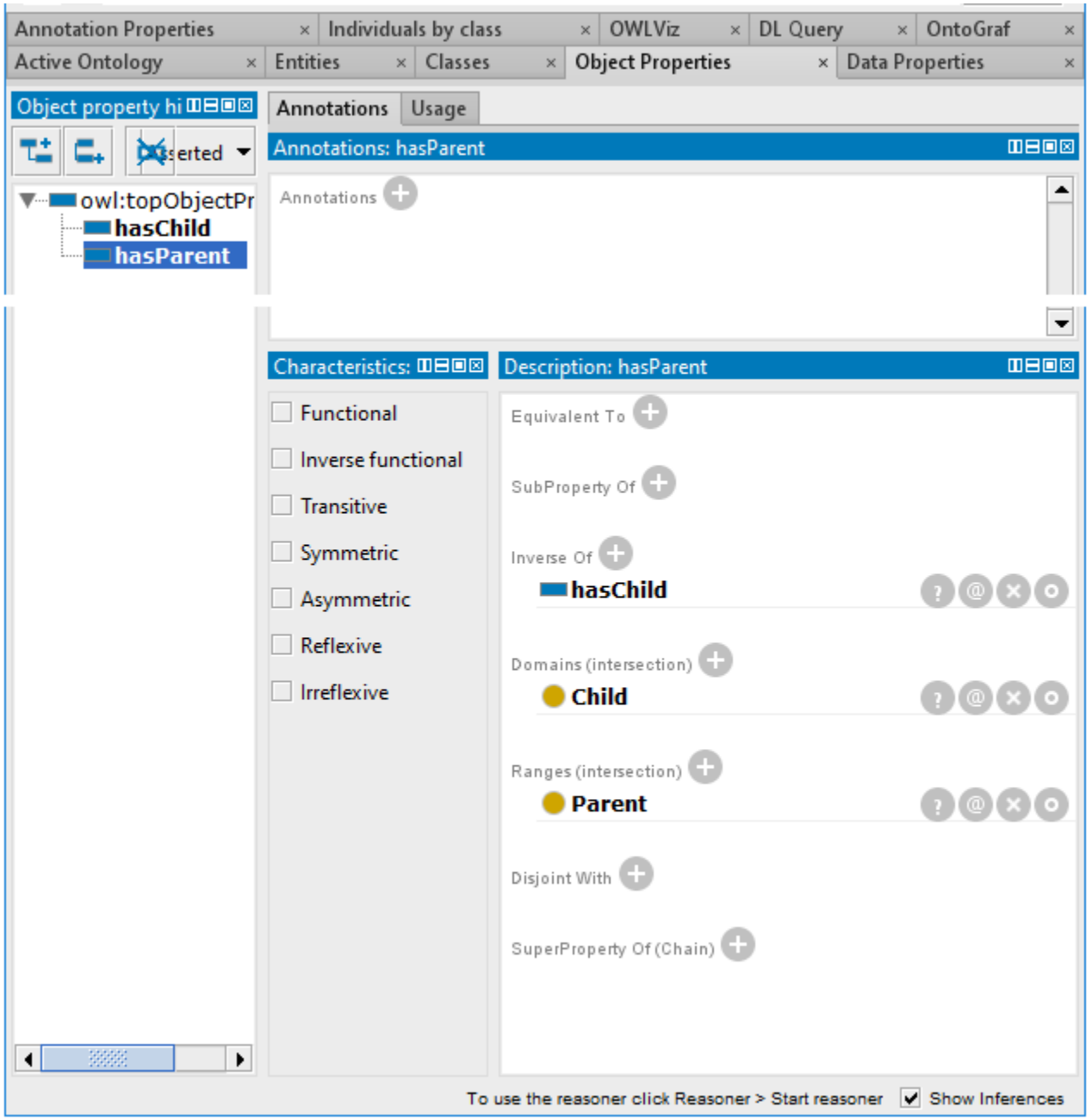
- individuals
- classes



Protégé

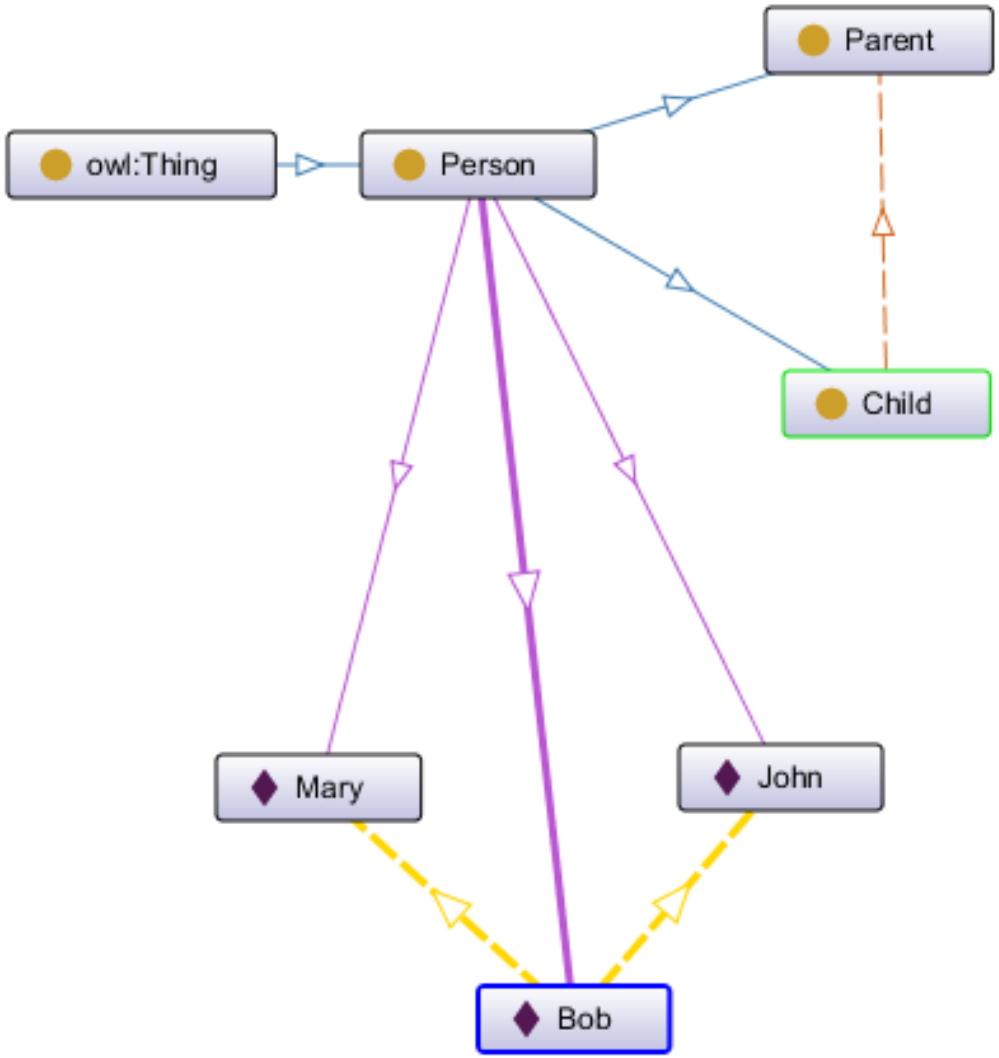
e.g.(2)

properties



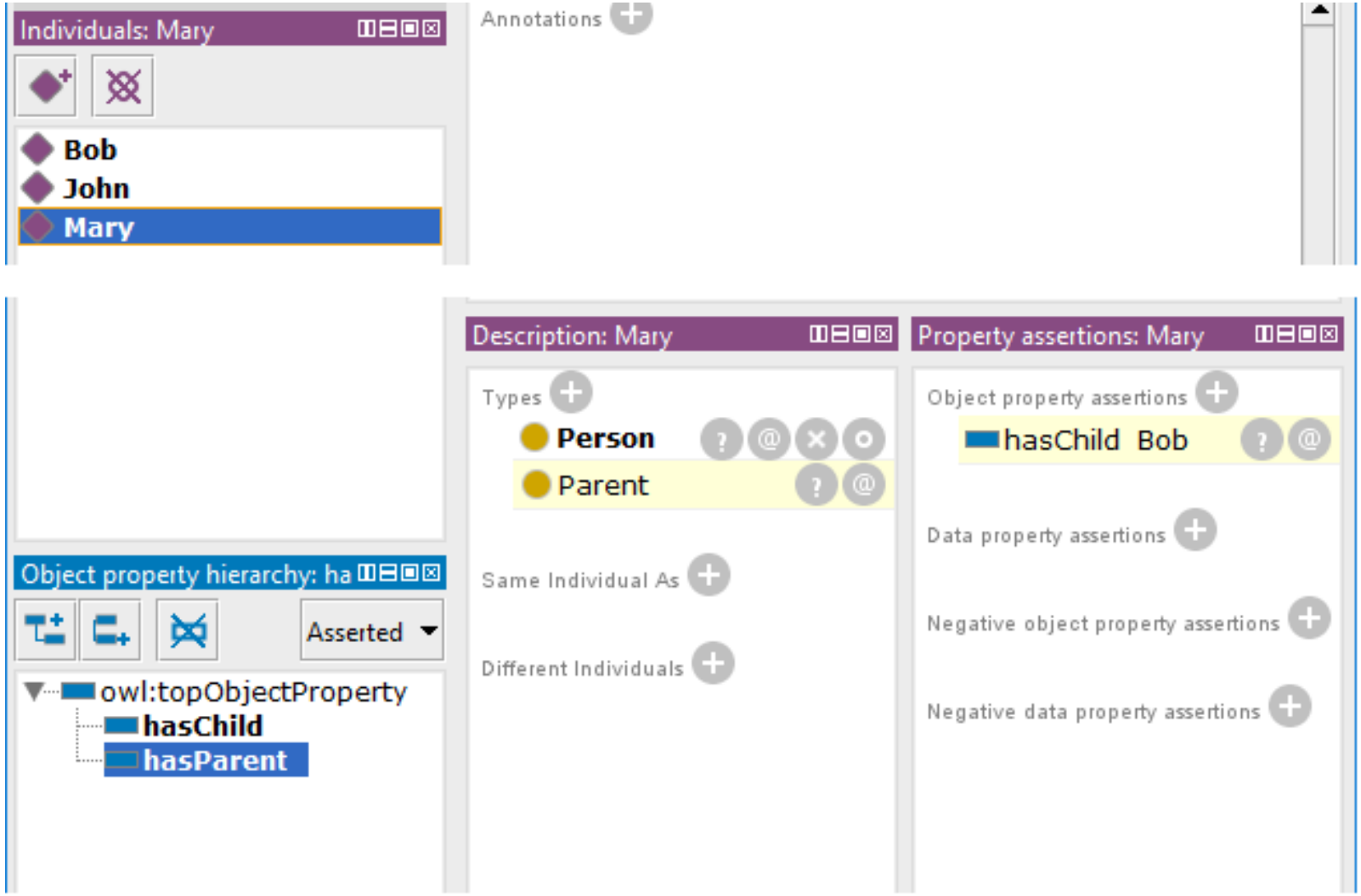
Protégé : OntoGraf

e.g.(2)



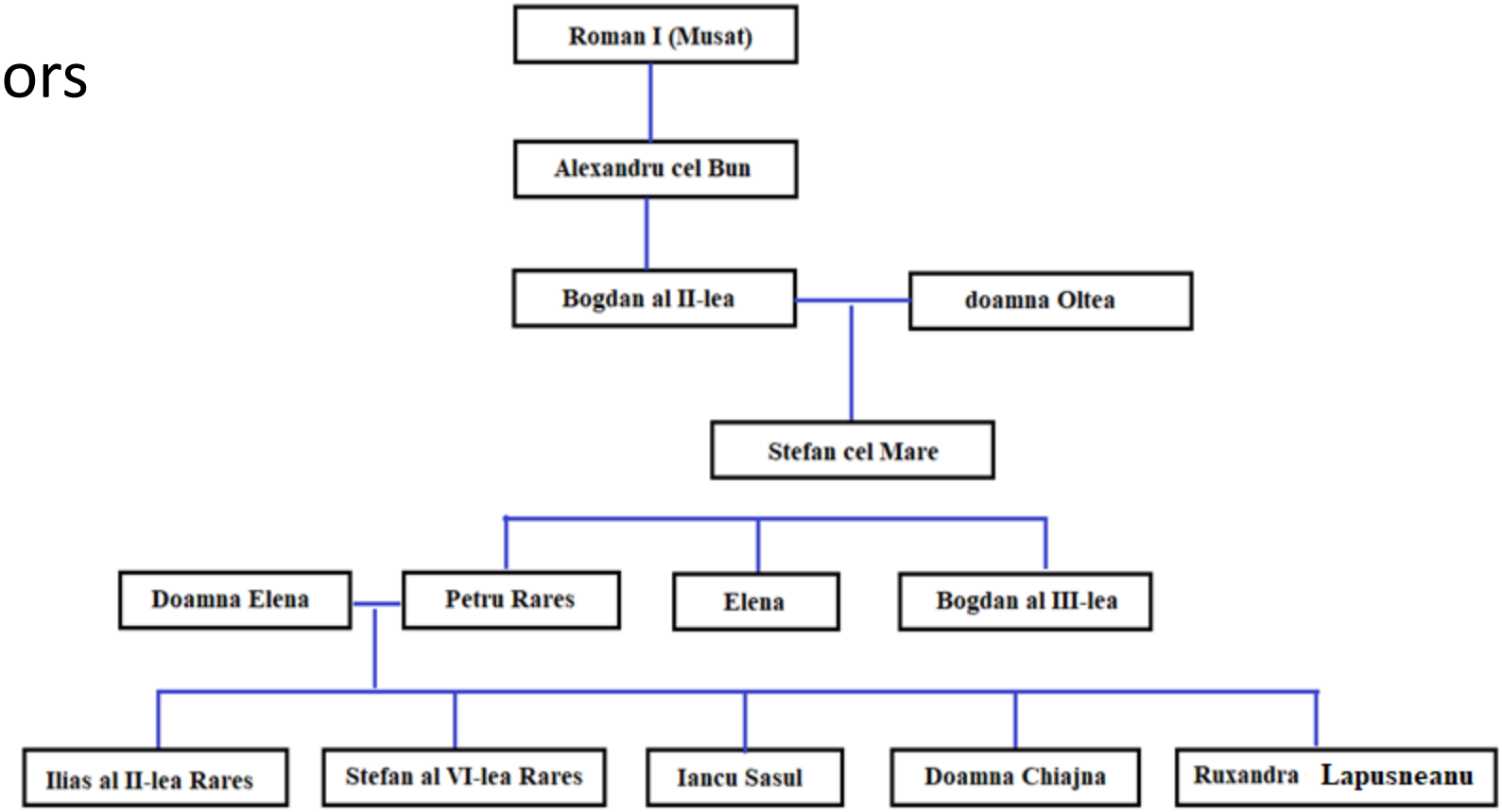
Protégé : Reasoner

e.g.(2)



Family example e.g. (3)

- Person
- Parent – Child relation
- Ancestors

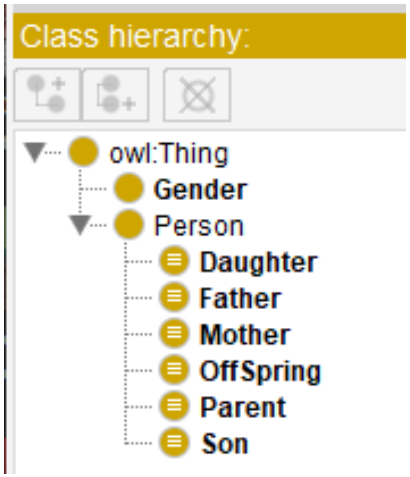


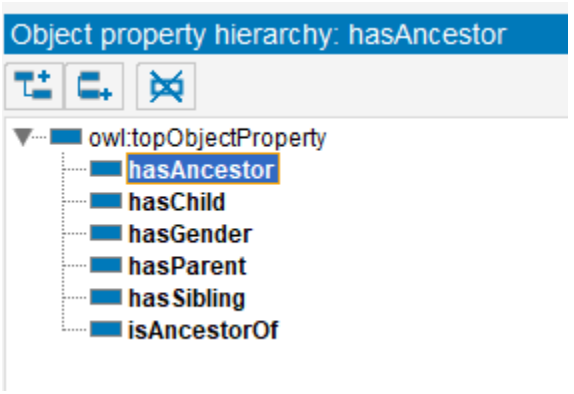
a short extract from the Musatini family (maybe not so accurate)

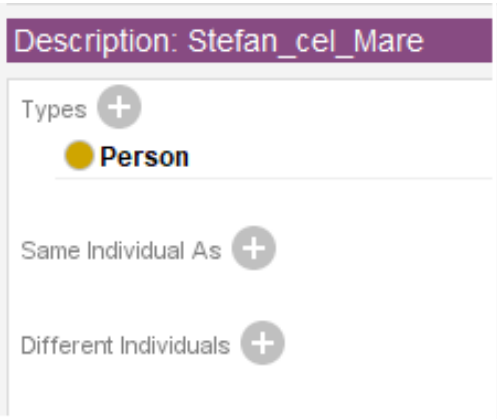
based on : https://ro.wikipedia.org/wiki/Dinastia_Musatinilor [accessed 2018]

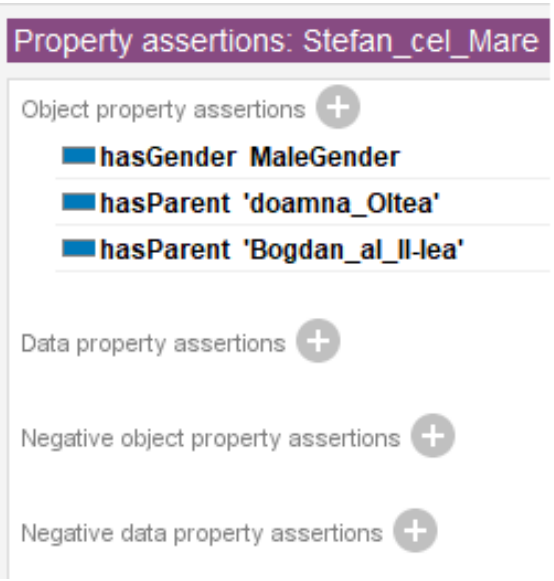
Family example e.g. (3)



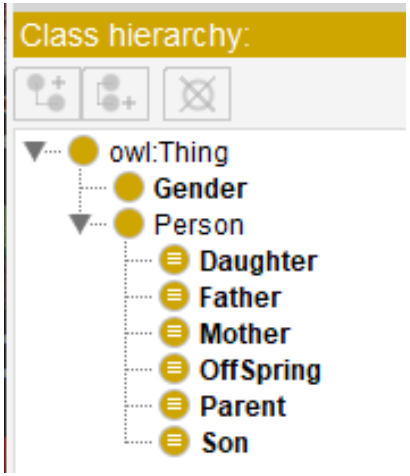


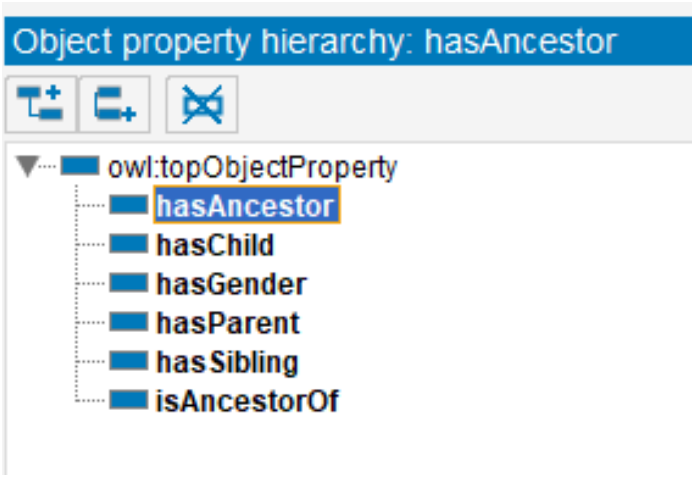


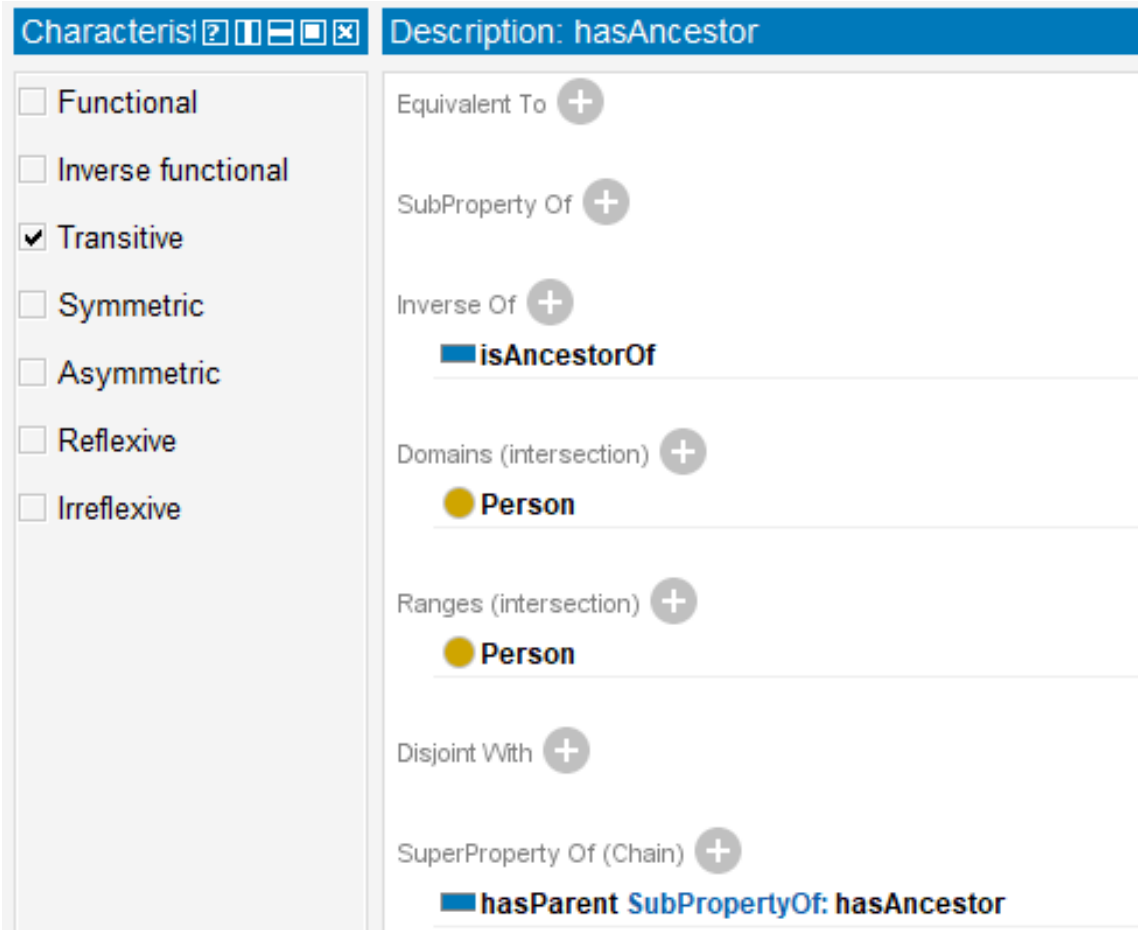




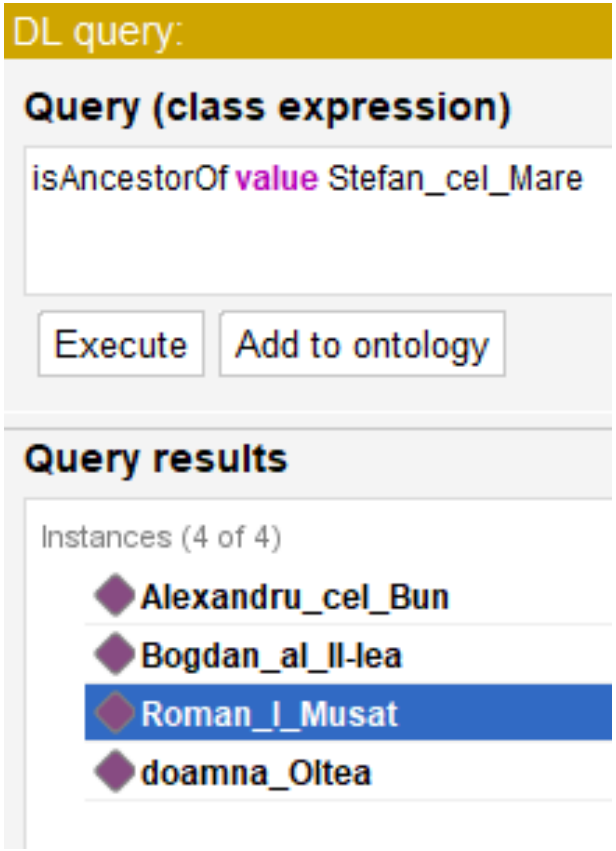
Family example e.g. (3)

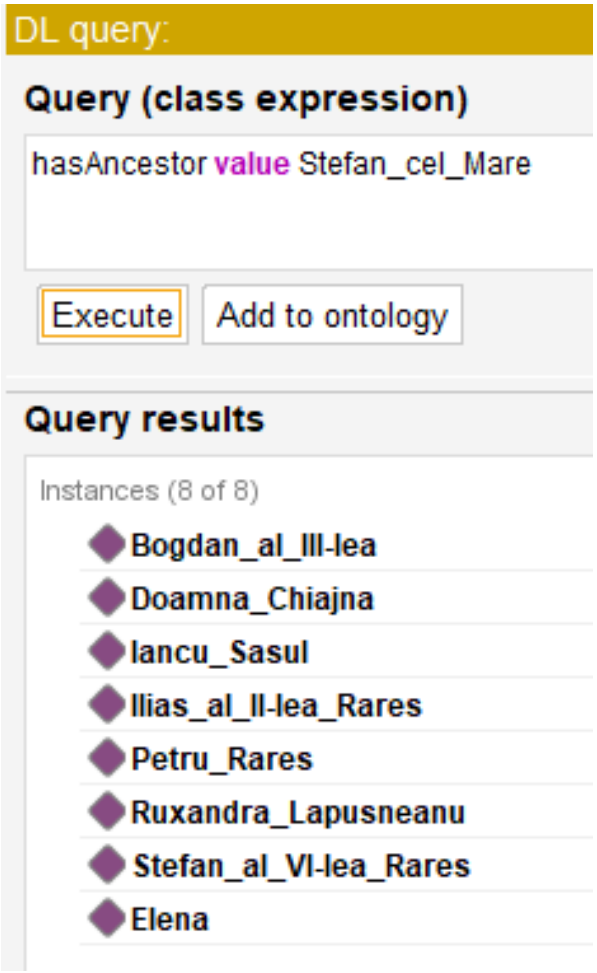






Family example e.g. (3)





Constraints

1. Quantifier restriction
   - at least a value from

     <has…> **some** <…> (values from)
   - all the values from

     <has…> **only** <…> (values from)
2. Cardinality restrictions

   <satisfy…> exactly (1,2,…) <…need>

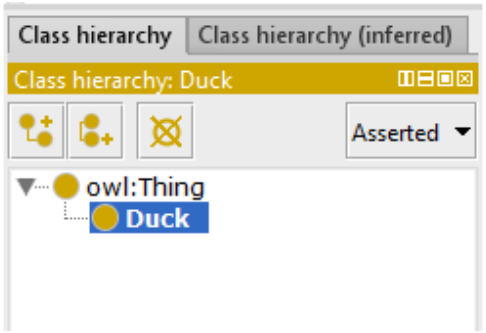
   max, min
3. <has…> **value**

e.g.(duck)

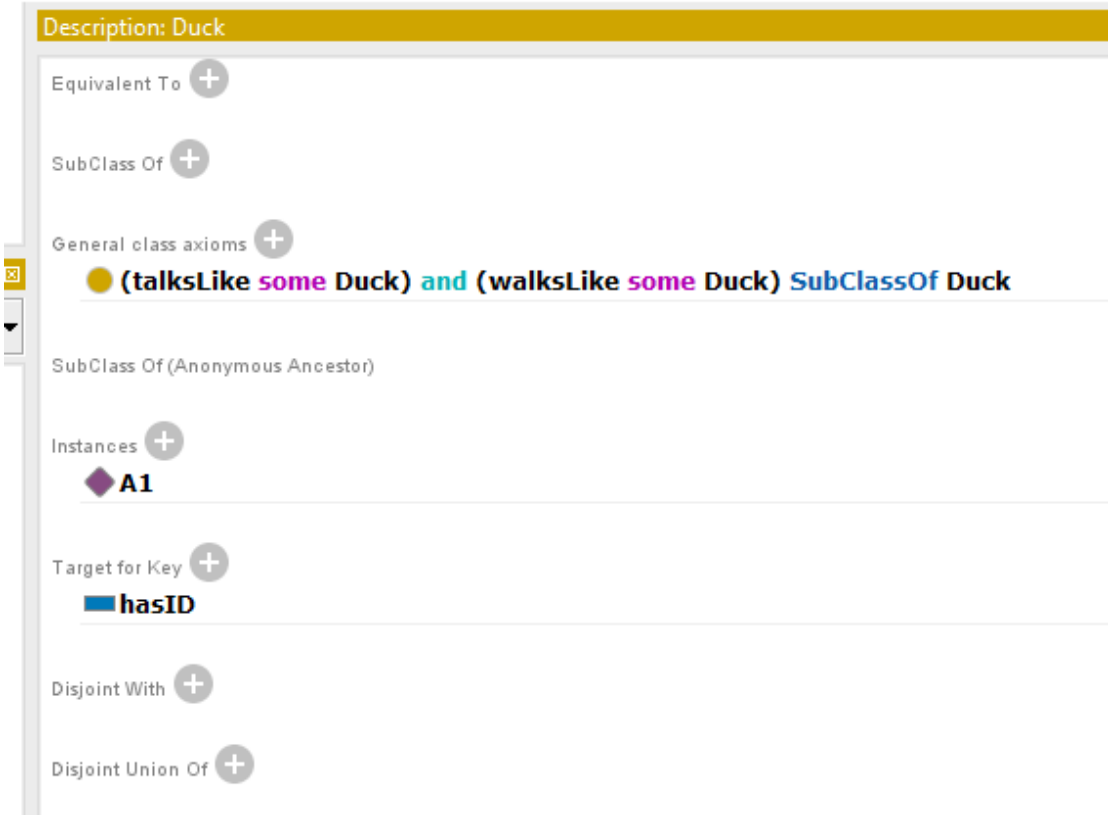
Write in Protégé:

“If it walks like a duck
and talks like a duck,
it is probably a duck”

e.g.(duck)



General class axiom example



Open world / Closed world

CWA (Closed World Assumption)

- a system has complete information

Anything that cannot be found is false



OWA (Open World Assumption)

- a system has incomplete information

Anything might be true unless it can be proven false

Guess what the Web is?

Open world / Closed world

- Negation as failure
- Negation as contradiction

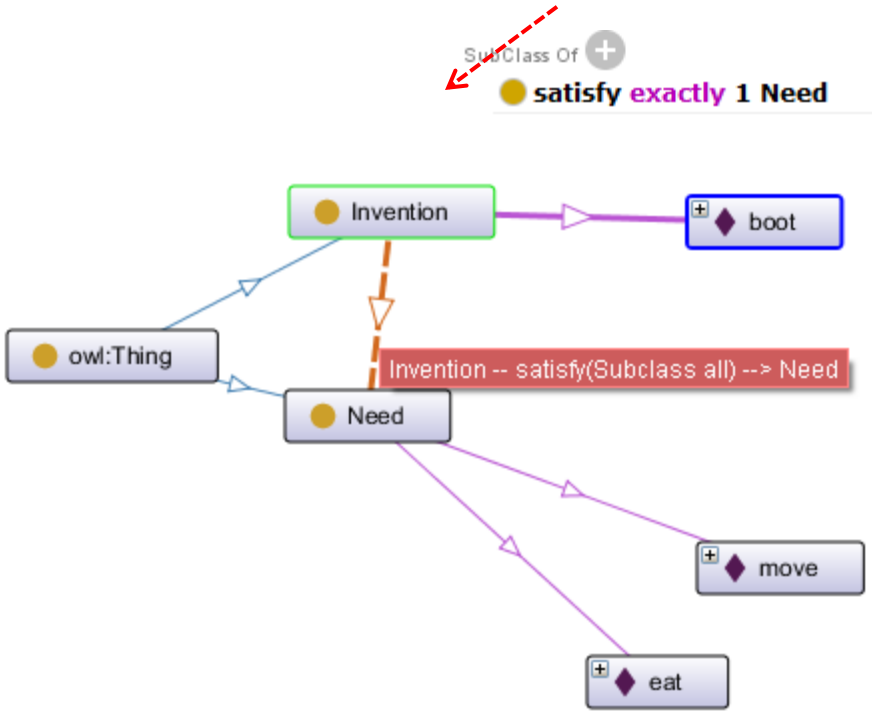
- OWA
- CWA

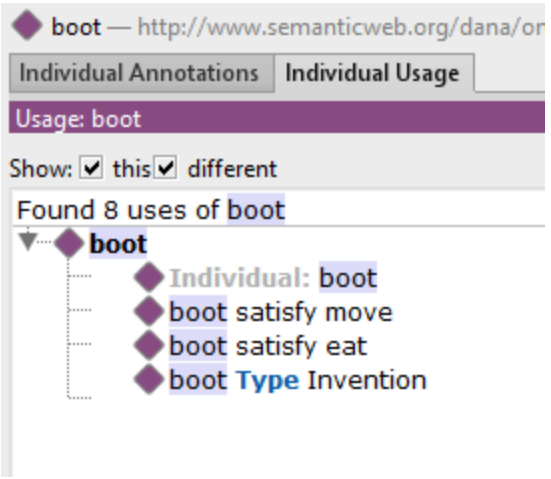


- Anything that cannot be found is possible

- Reasoning about any world consistent with the model

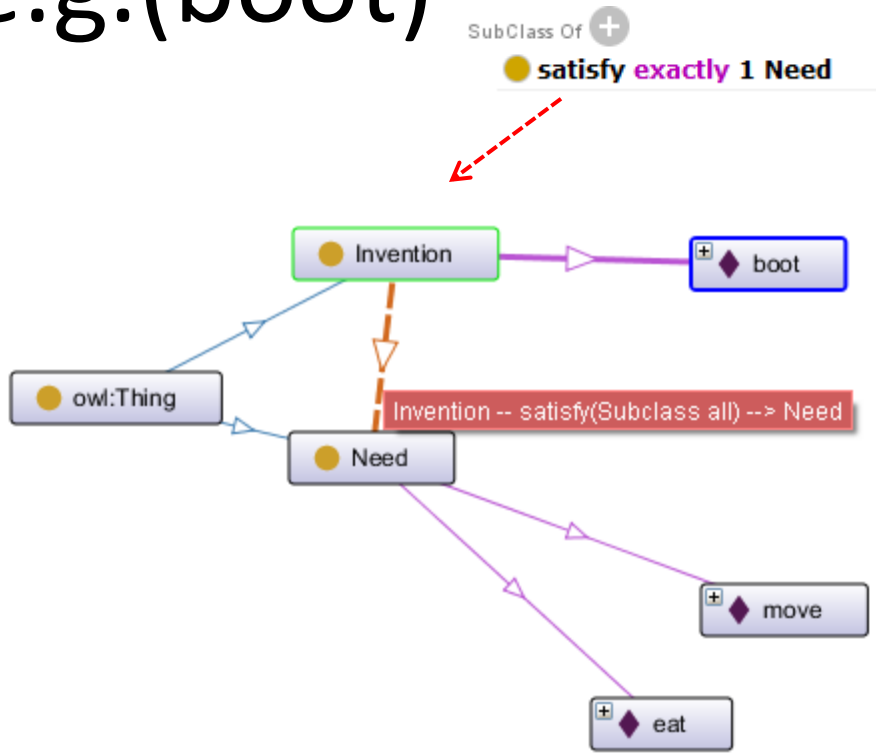
E.g. (boot) : “An invention satisfies exactly one need”

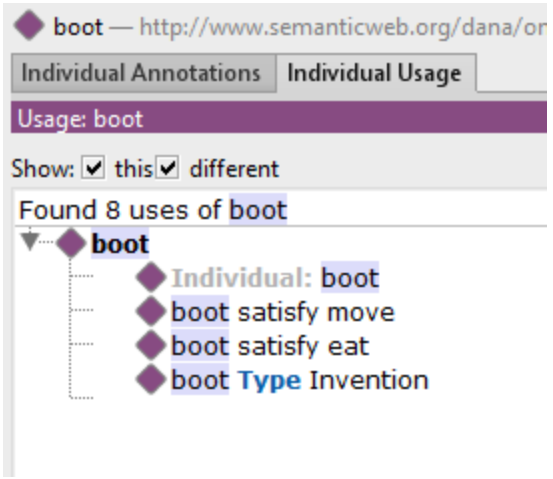




… everything OK ?

e.g.(boot)

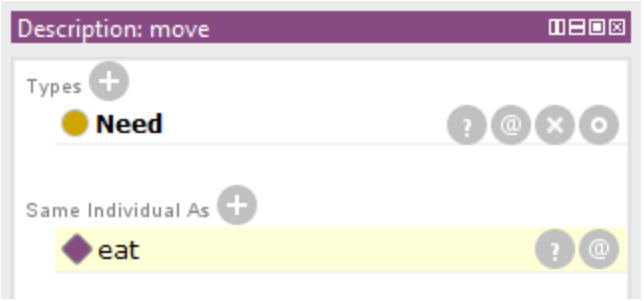




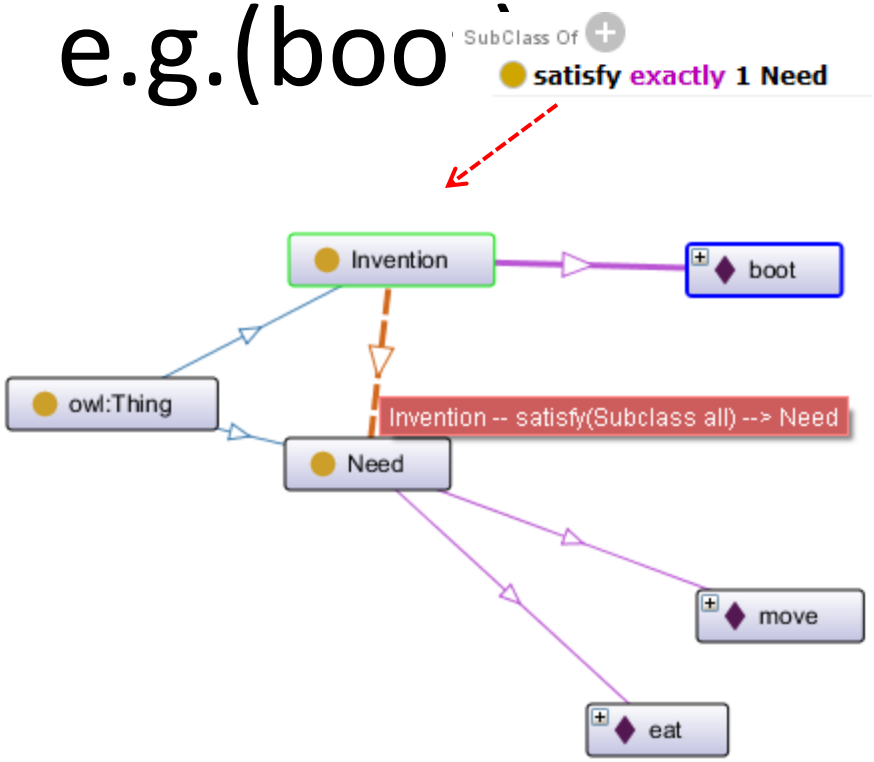
… everything OK ?

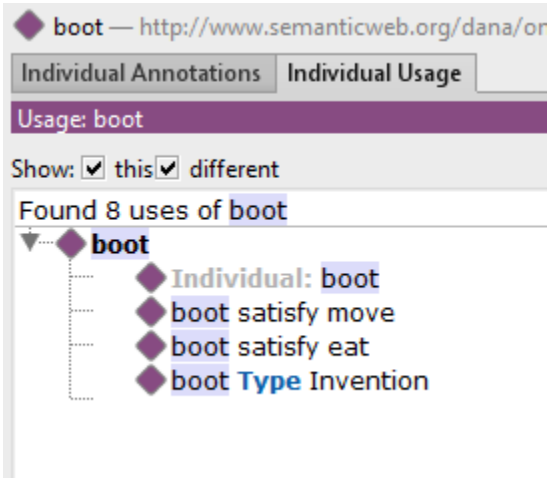
Start REASONER!

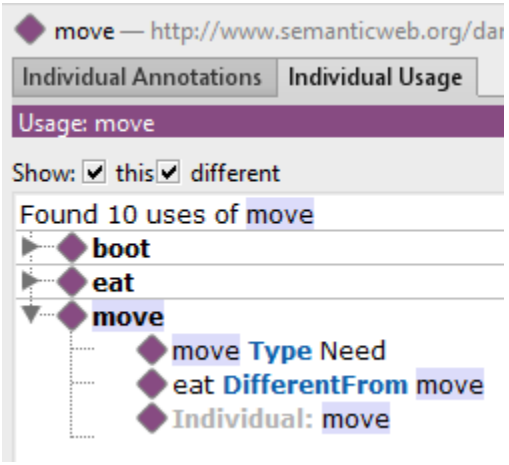
individual move:



e.g.(boo

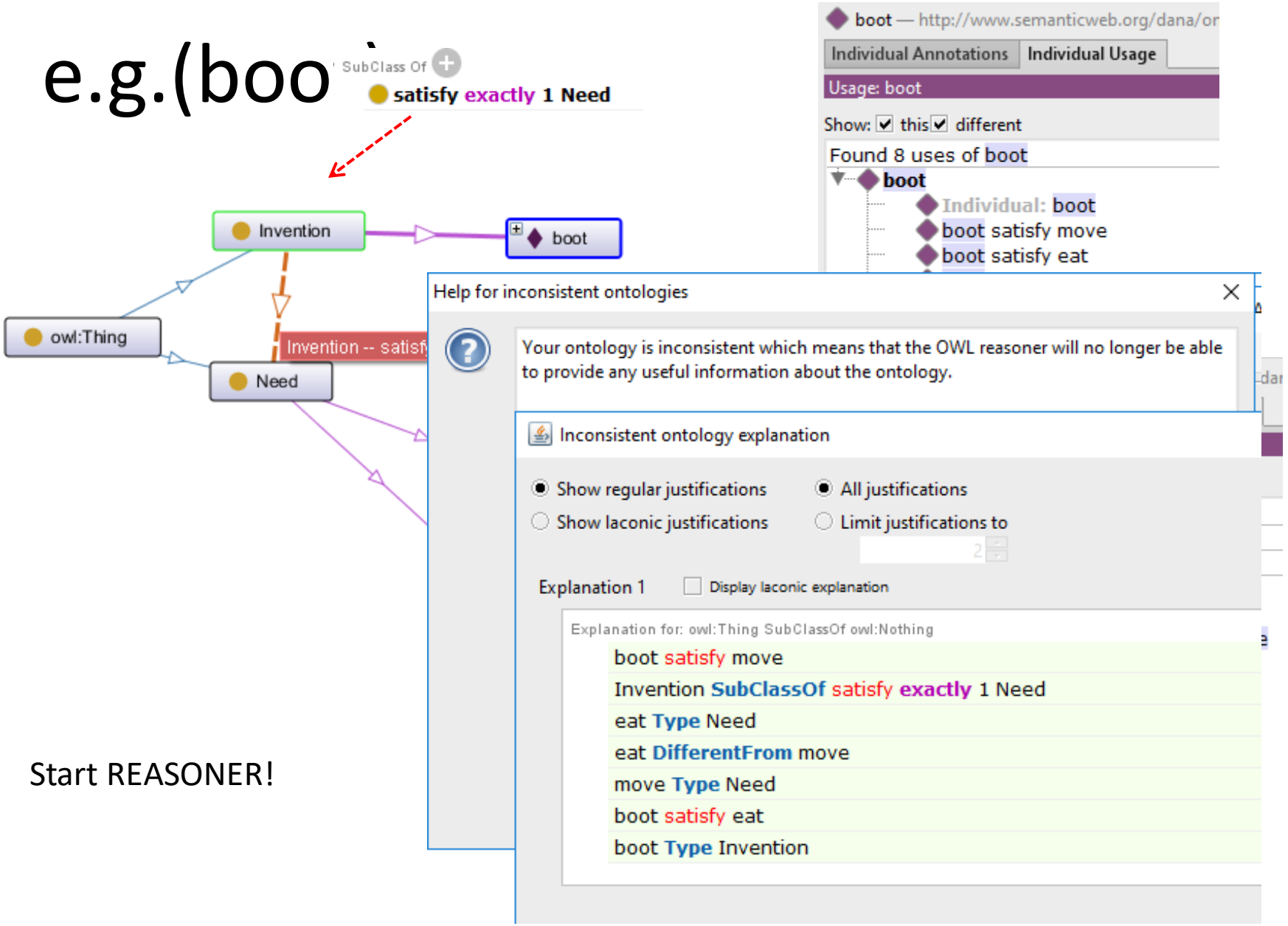






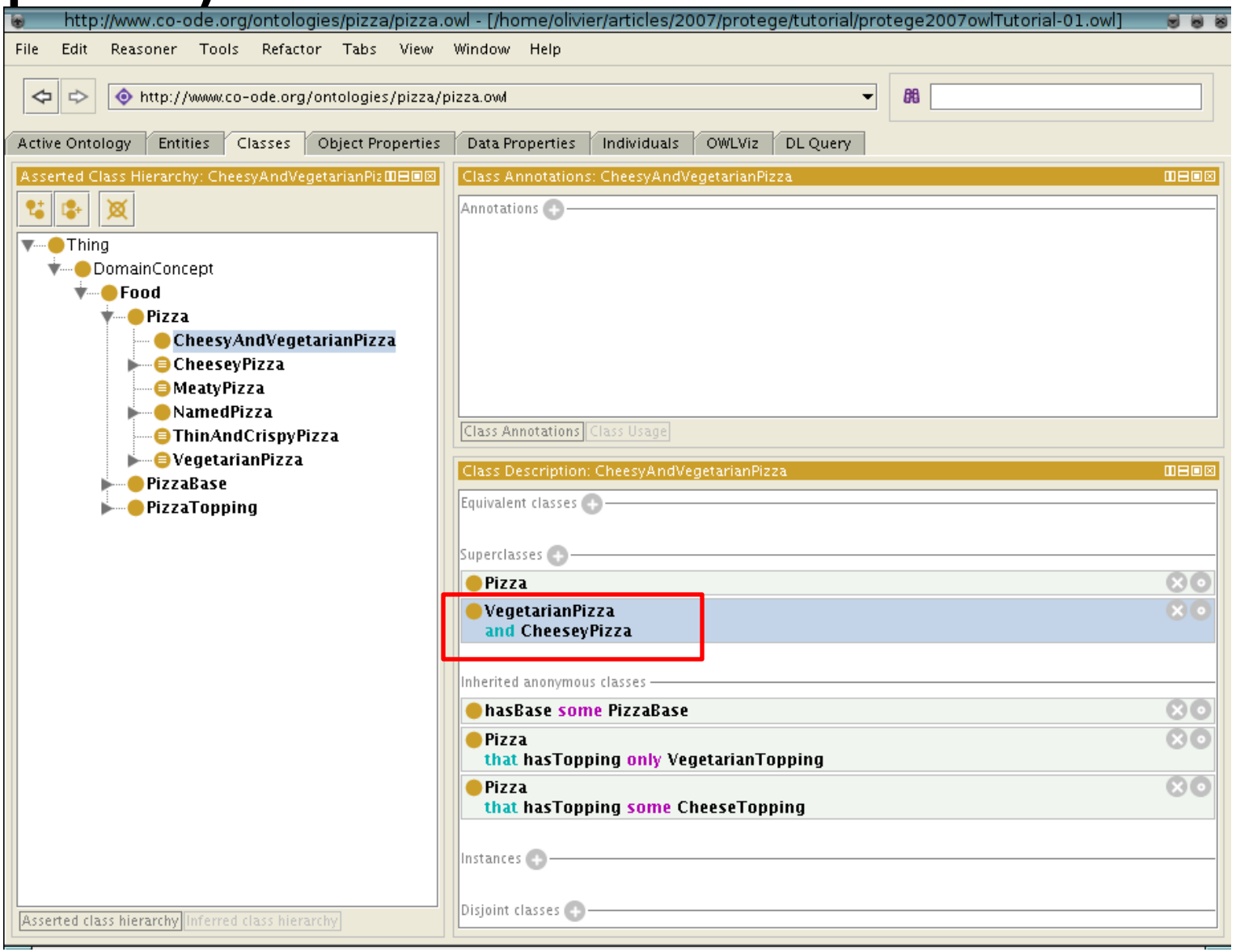
Everything it is OK ?

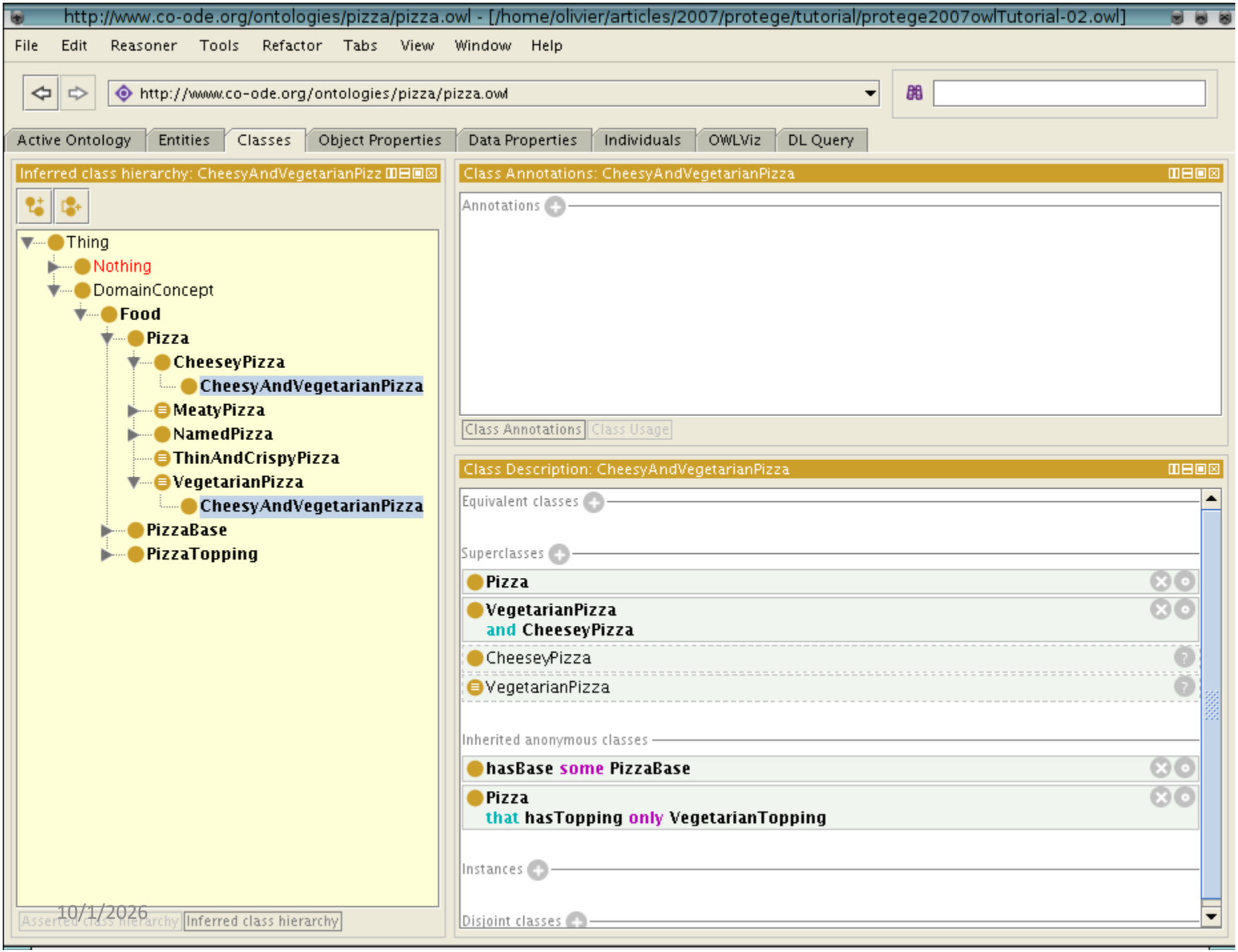
e.g.(boo

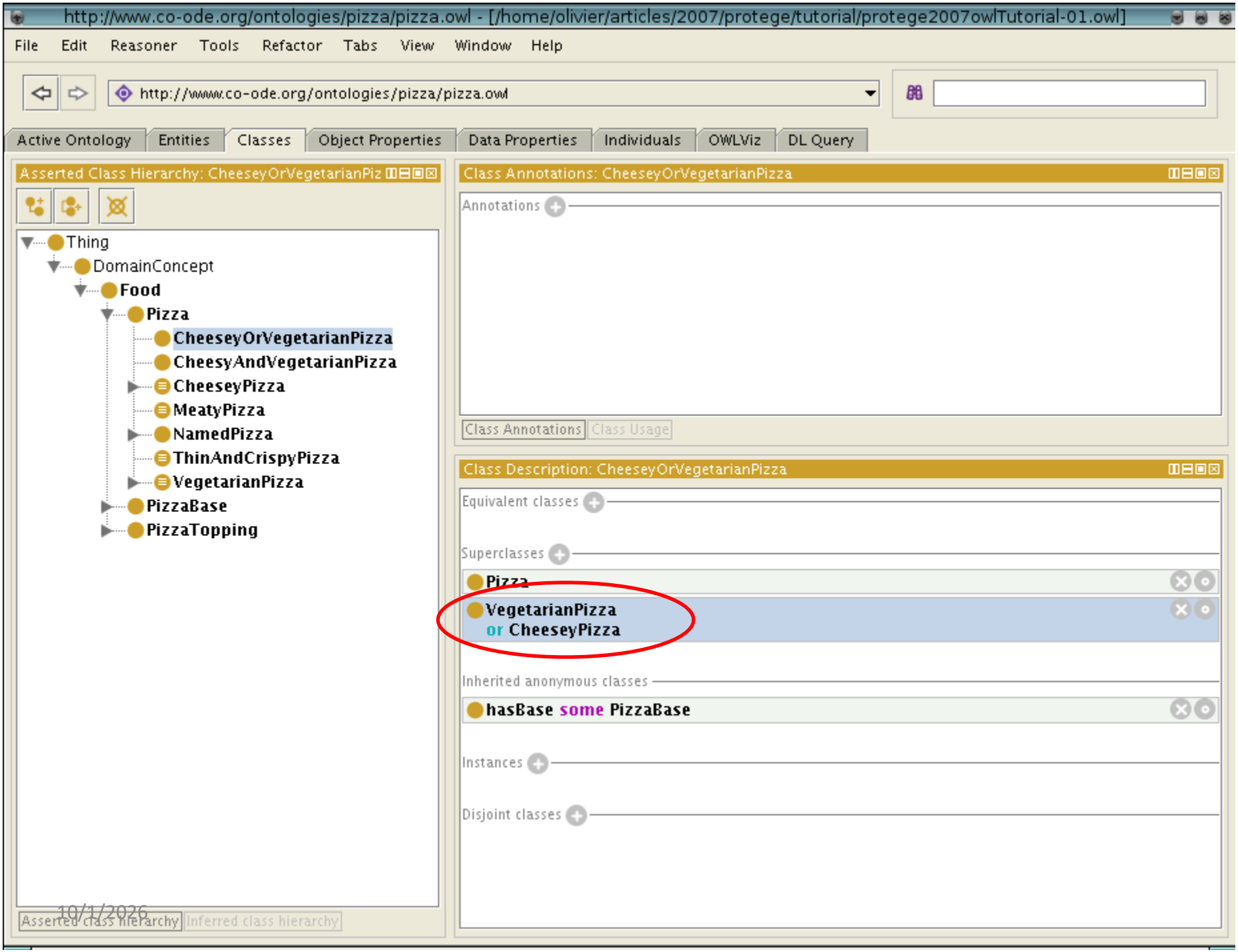


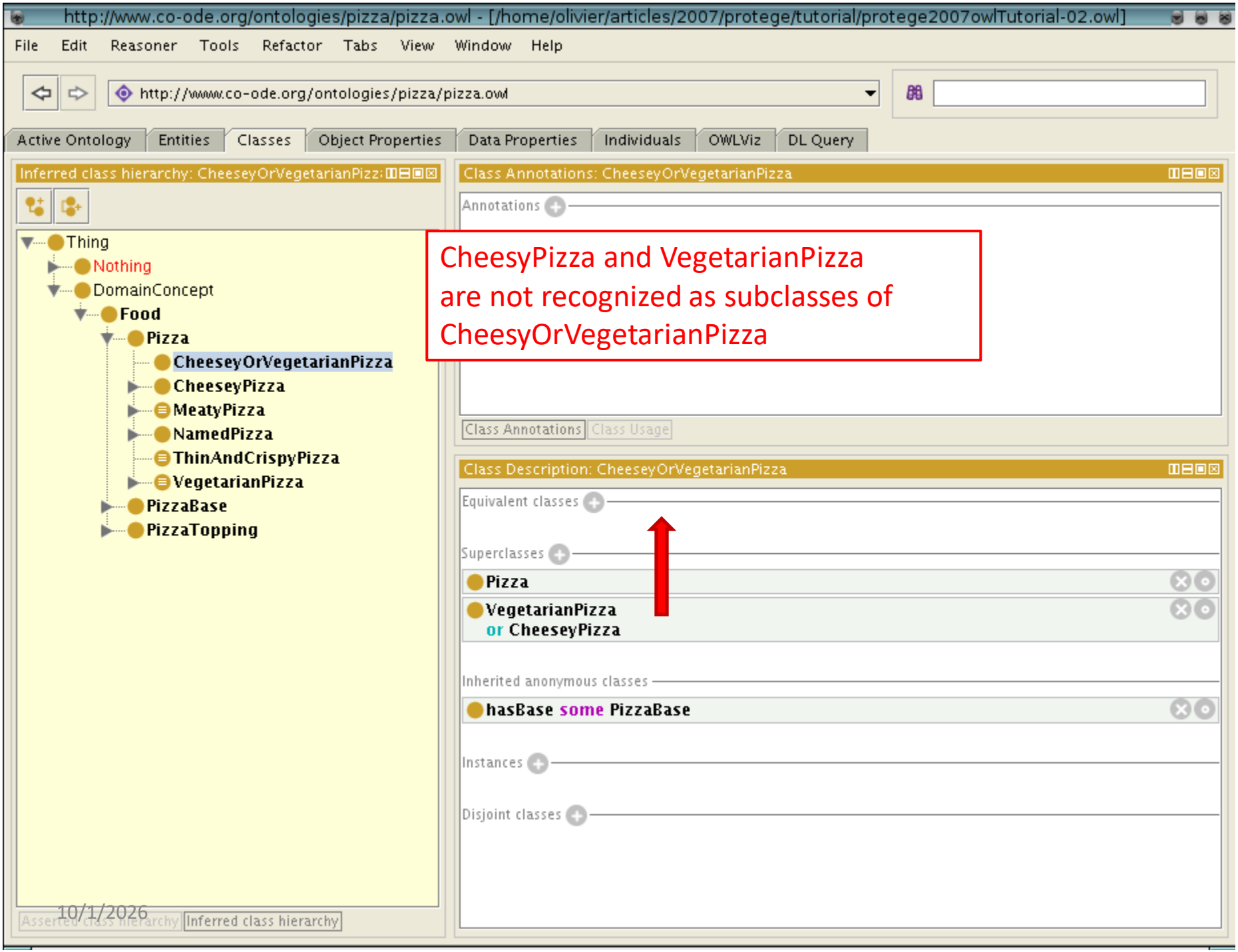
Start REASONER!

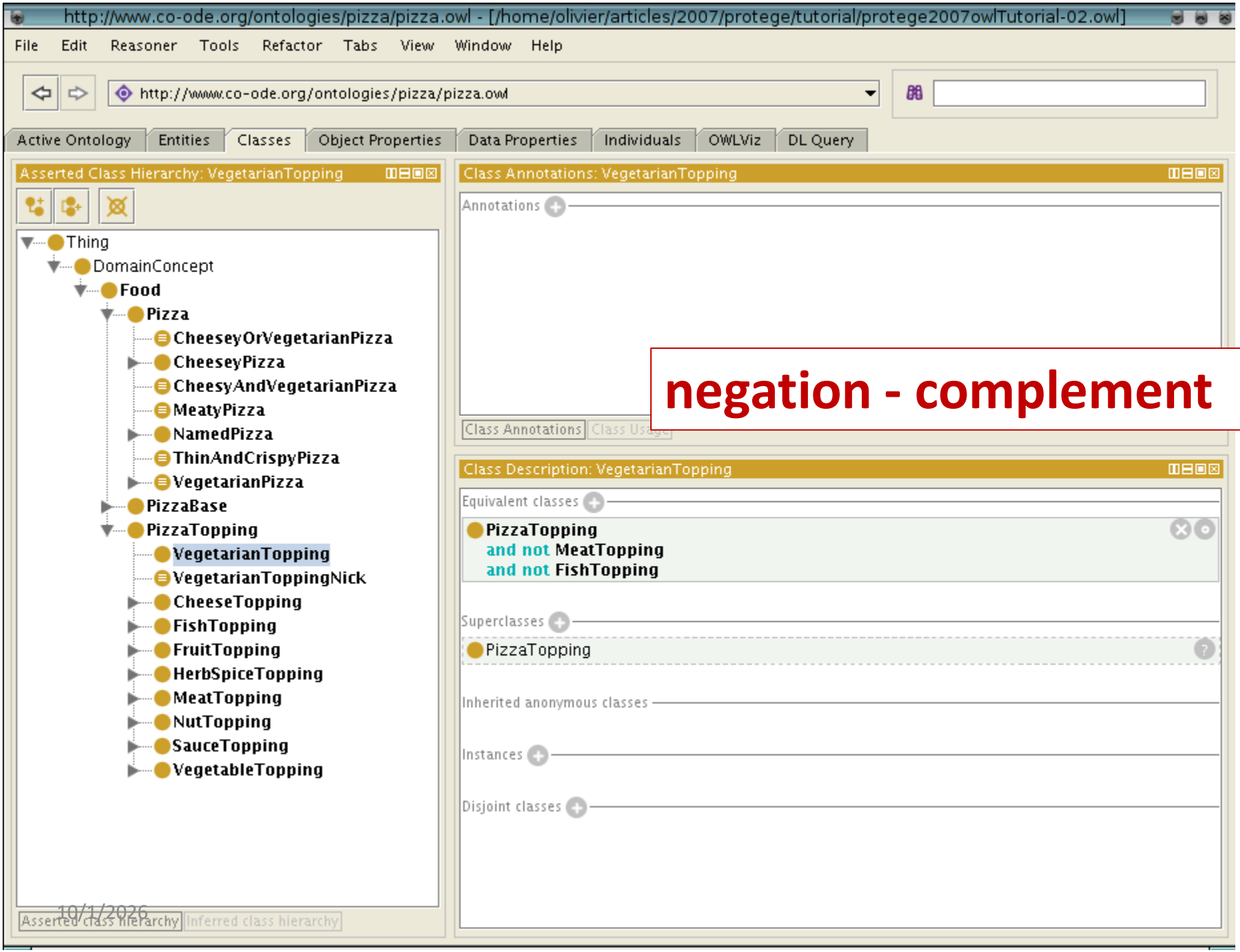
e.g.(4pizza)

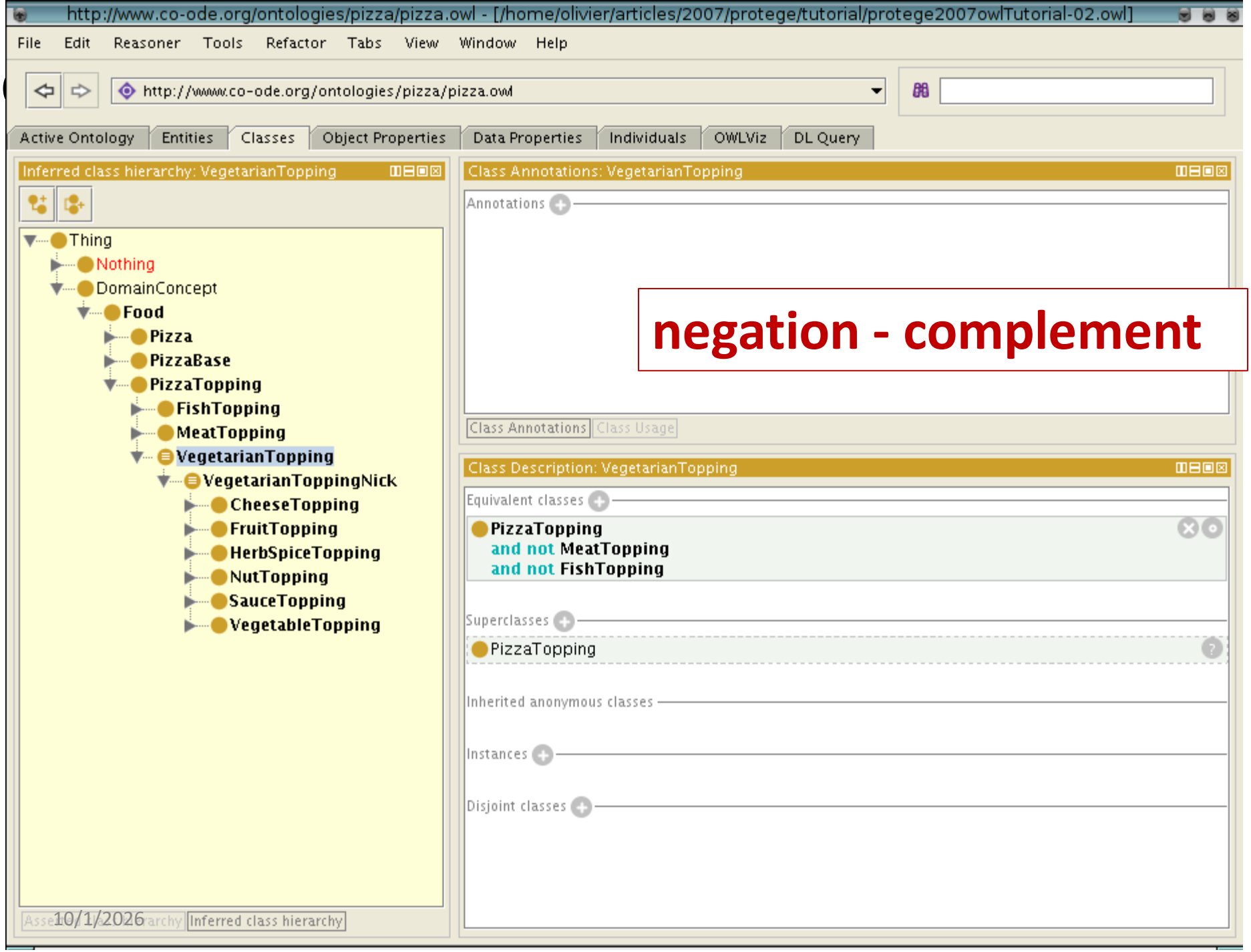

http://www.co-ode.org/ontologies/pizza/pizza.owl - [/home/olivier/articles/2007/protege/tutorial/protege2007owlTutorial-01.owl]
File Edit Reasoner Tools Refactor Tabs View Window Help
http://www.co-ode.org/ontologies/pizza/pizza.owl
Active Ontology Entities Classes Object Properties Data Properties Individuals OWLViz DL Query
Asserted Class Hierarchy: CheesyAndVegetarianPiz
Thing
DomainConcept
Food
Pizza
CheesyAndVegetarianPizza
CheeseyPizza
MeatyPizza
NamedPizza
ThinAndCrispyPizza
VegetarianPizza
PizzaBase
PizzaTopping
Asserted class hierarchy
Inferred class hierarchy
Class Annotations: CheesyAndVegetarianPizza
Annotations
Class Annotations
Class Usage
Class Description: CheesyAndVegetarianPizza
Equivalent classes
Superclasses
Pizza
VegetarianPizza
and CheeseyPizza
Inherited anonymous classes
hasBase some PizzaBase
Pizza
that hasTopping only VegetarianTopping
Pizza
that hasTopping some CheeseTopping
Instances
Disjoint classes

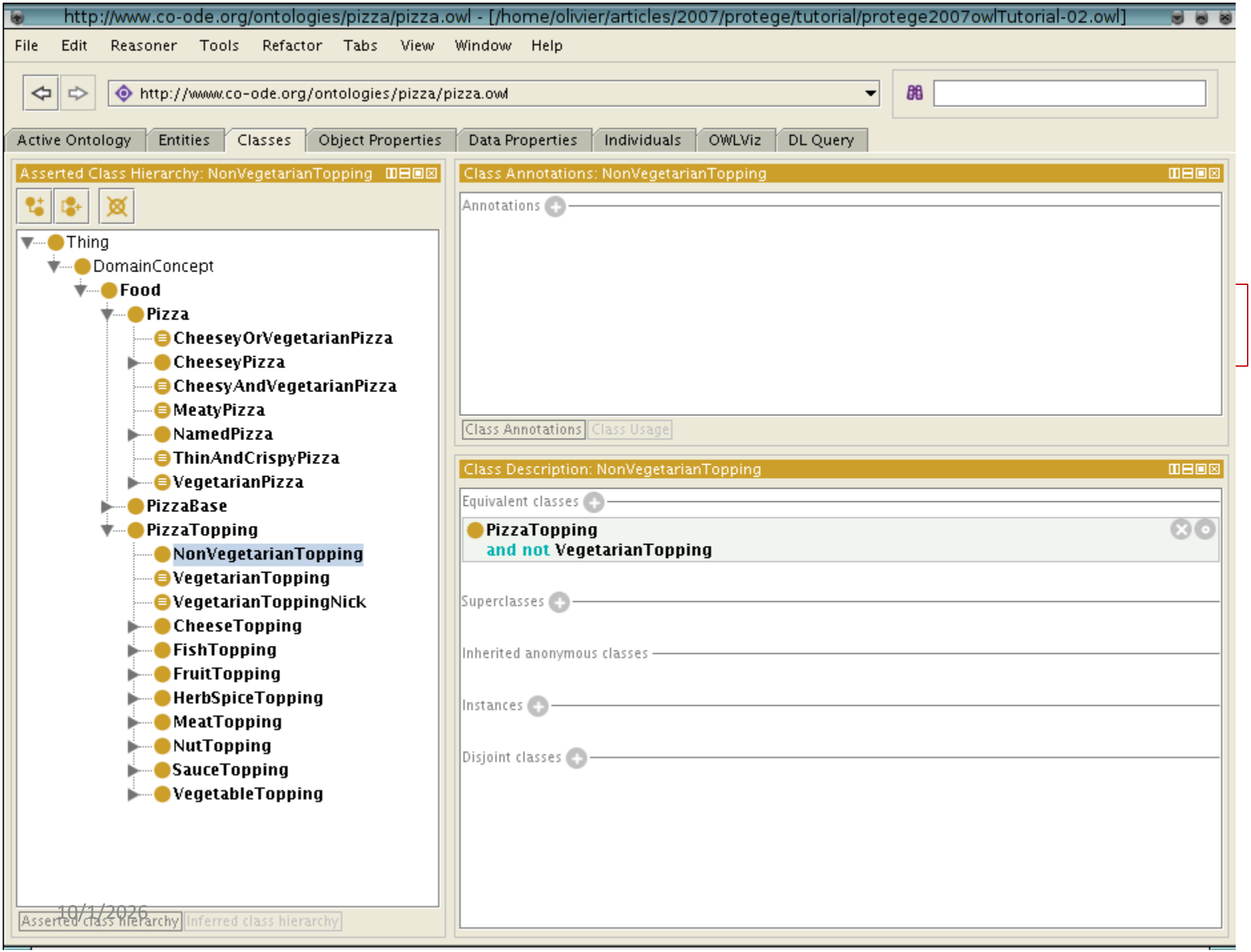

http://www.co-ode.org/ontologies/pizza/pizza.owl - [/home/olivier/articles/2007/protege/tutorial/protege2007owlTutorial-02.owl]
File Edit Reasoner Tools Refactor Tabs View Window Help
http://www.co-ode.org/ontologies/pizza/pizza.owl
Active Ontology Entities Classes Object Properties Data Properties Individuals OWLViz DL Query
Inferred class hierarchy: CheesyAndVegetarianPizz
Thing
Nothing
DomainConcept
Food
Pizza
CheeseyPizza
CheesyAndVegetarianPizza
MeatyPizza
NamedPizza
ThinAndCrispyPizza
VegetarianPizza
CheesyAndVegetarianPizza
PizzaBase
PizzaTopping
Asserted class hierarchy
Inferred class hierarchy
Class Annotations: CheesyAndVegetarianPizza
Annotations
Class Annotations
Class Usage
Class Description: CheesyAndVegetarianPizza
Equivalent classes
Superclasses
Pizza
VegetarianPizza
and CheeseyPizza
CheeseyPizza
VegetarianPizza
Inherited anonymous classes
hasBase some PizzaBase
Pizza
that hasTopping only VegetarianTopping
Instances
Disjoint classes

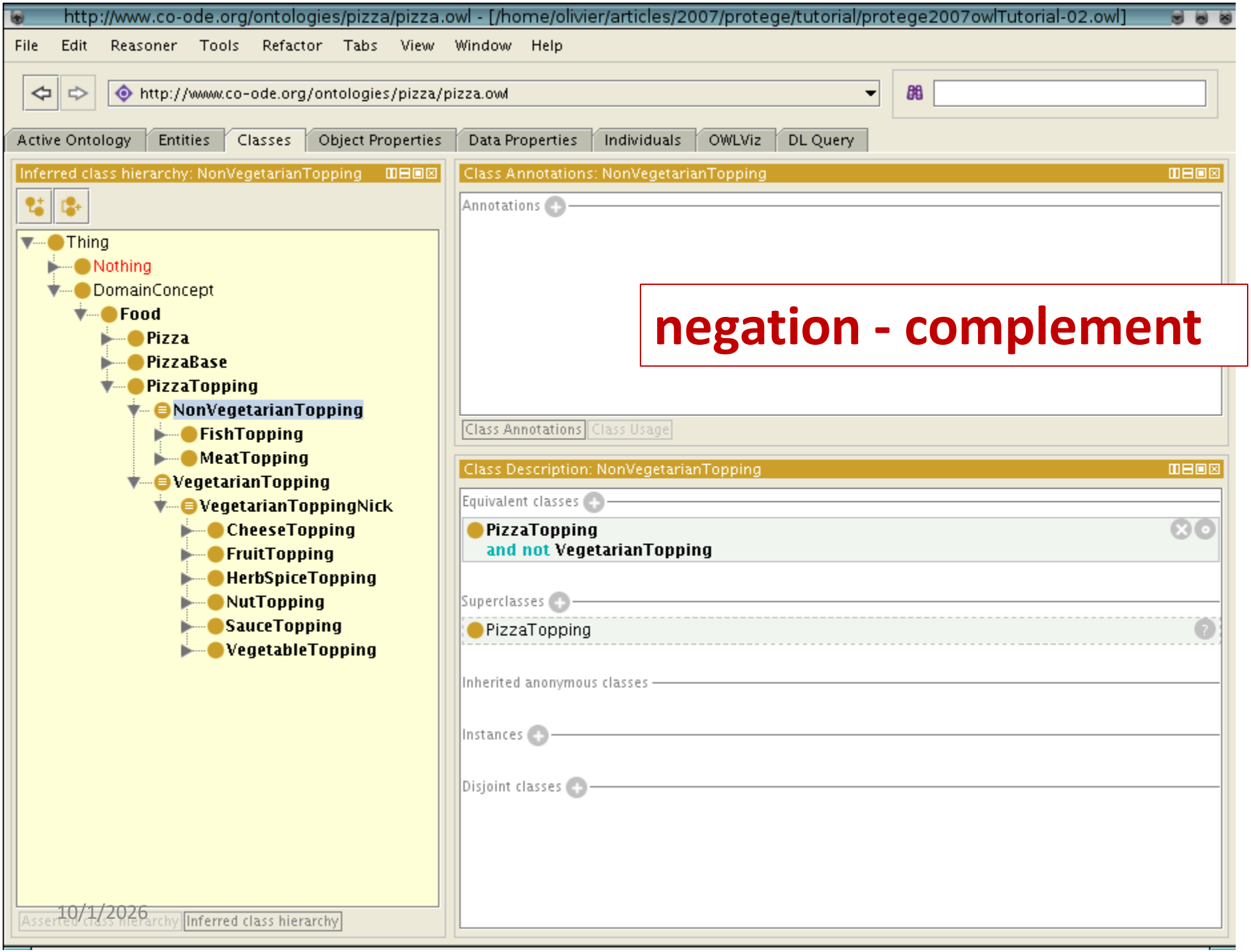

http://www.co-ode.org/ontologies/pizza/pizza.owl - [/home/olivier/articles/2007/protege/tutorial/protege2007owlTutorial-01.owl]
File Edit Reasoner Tools Refactor Tabs View Window Help
http://www.co-ode.org/ontologies/pizza/pizza.owl
Active Ontology Entities Classes Object Properties Data Properties Individuals OWLViz DL Query
Asserted Class Hierarchy: CheeseyOrVegetarianPiz
Thing
DomainConcept
Food
Pizza
CheeseyOrVegetarianPizza
CheesyAndVegetarianPizza
CheeseyPizza
MeatyPizza
NamedPizza
ThinAndCrispyPizza
VegetarianPizza
PizzaBase
PizzaTopping
Asserted class hierarchy
Inferred class hierarchy
Class Annotations: CheeseyOrVegetarianPizza
Annotations
Class Annotations
Class Usage
Class Description: CheeseyOrVegetarianPizza
Equivalent classes
Superclasses
Pizza
VegetarianPizza
or CheeseyPizza
Inherited anonymous classes
hasBase some PizzaBase
Instances
Disjoint classes


http://www.co-ode.org/ontologies/pizza/pizza.owl - [/home/olivier/articles/2007/protege/tutorial/protege2007owlTutorial-02.owl]
File Edit Reasoner Tools Refactor Tabs View Window Help
http://www.co-ode.org/ontologies/pizza/pizza.owl
Active Ontology Entities Classes Object Properties Data Properties Individuals OWLViz DL Query
Inferred class hierarchy: CheeseyOrVegetarianPizz:
Thing
Nothing
DomainConcept
Food
Pizza
CheeseyOrVegetarianPizza
CheeseyPizza
MeatyPizza
NamedPizza
ThinAndCrispyPizza
VegetarianPizza
PizzaBase
PizzaTopping
Asserted class hierarchy
Inferred class hierarchy
Class Annotations: CheeseyOrVegetarianPizza
Annotations
Class Annotations
Class Usage
CheesyPizza and VegetarianPizza
are not recognized as subclasses of
CheesyOrVegetarianPizza
Class Description: CheeseyOrVegetarianPizza
Equivalent classes
Superclasses
Pizza
VegetarianPizza
or CheeseyPizza
Inherited anonymous classes
hasBase some PizzaBase
Instances
Disjoint classes


http://www.co-ode.org/ontologies/pizza/pizza.owl - [/home/olivier/articles/2007/protege/tutorial/protege2007owlTutorial-02.owl]
File Edit Reasoner Tools Refactor Tabs View Window Help
http://www.co-ode.org/ontologies/pizza/pizza.owl
Active Ontology Entities Classes Object Properties Data Properties Individuals OWLViz DL Query
Asserted Class Hierarchy: VegetarianTopping
Thing
DomainConcept
Food
Pizza
CheeseyOrVegetarianPizza
CheeseyPizza
CheesyAndVegetarianPizza
MeatyPizza
NamedPizza
ThinAndCrispyPizza
VegetarianPizza
PizzaBase
PizzaTopping
VegetarianTopping
VegetarianToppingNick
CheeseTopping
FishTopping
FruitTopping
HerbSpiceTopping
MeatTopping
NutTopping
SauceTopping
VegetableTopping
Asserted class hierarchy
Inferred class hierarchy
Class Annotations: VegetarianTopping
Annotations
Class Annotations
Class Usage
negation - complement
Class Description: VegetarianTopping
Equivalent classes
PizzaTopping
and not MeatTopping
and not FishTopping
Superclasses
PizzaTopping
Inherited anonymous classes
Instances
Disjoint classes


http://www.co-ode.org/ontologies/pizza/pizza.owl - [/home/olivier/articles/2007/protege/tutorial/protege2007owlTutorial-02.owl]
File Edit Reasoner Tools Refactor Tabs View Window Help
http://www.co-ode.org/ontologies/pizza/pizza.owl
Active Ontology Entities Classes Object Properties Data Properties Individuals OWLViz DL Query
Inferred class hierarchy: VegetarianTopping
Thing
Nothing
DomainConcept
Food
Pizza
PizzaBase
PizzaTopping
FishTopping
MeatTopping
VegetarianTopping
VegetarianToppingNick
CheeseTopping
FruitTopping
HerbSpiceTopping
NutTopping
SauceTopping
VegetableTopping
Inferred class hierarchy
Class Annotations: VegetarianTopping
Annotations
Class Annotations
Class Usage
negation - complement
Class Description: VegetarianTopping
Equivalent classes
PizzaTopping
and not MeatTopping
and not FishTopping
Superclasses
PizzaTopping
Inherited anonymous classes
Instances
Disjoint classes


http://www.co-ode.org/ontologies/pizza/pizza.owl - [/home/olivier/articles/2007/protege/tutorial/protege2007owlTutorial-02.owl]
File Edit Reasoner Tools Refactor Tabs View Window Help
http://www.co-ode.org/ontologies/pizza/pizza.owl
Active Ontology Entities Classes Object Properties Data Properties Individuals OWLViz DL Query
Asserted Class Hierarchy: NonVegetarianTopping
Thing
DomainConcept
Food
Pizza
CheeseyOrVegetarianPizza
CheeseyPizza
CheesyAndVegetarianPizza
MeatyPizza
NamedPizza
ThinAndCrispyPizza
VegetarianPizza
PizzaBase
PizzaTopping
NonVegetarianTopping
VegetarianTopping
VegetarianToppingNick
CheeseTopping
FishTopping
FruitTopping
HerbSpiceTopping
MeatTopping
NutTopping
SauceTopping
VegetableTopping
Asserted class hierarchy
Inferred class hierarchy
Class Annotations: NonVegetarianTopping
Annotations
Class Annotations
Class Usage
Class Description: NonVegetarianTopping
Equivalent classes
PizzaTopping
and not VegetarianTopping
Superclasses
Inherited anonymous classes
Instances
Disjoint classes


http://www.co-ode.org/ontologies/pizza/pizza.owl - [/home/olivier/articles/2007/protege/tutorial/protege2007owlTutorial-02.owl]
File Edit Reasoner Tools Refactor Tabs View Window Help
http://www.co-ode.org/ontologies/pizza/pizza.owl
Active Ontology Entities Classes Object Properties Data Properties Individuals OWLViz DL Query
Inferred class hierarchy: NonVegetarianTopping
Thing
Nothing
DomainConcept
Food
Pizza
PizzaBase
PizzaTopping
NonVegetarianTopping
FishTopping
MeatTopping
VegetarianTopping
VegetarianToppingNick
CheeseTopping
FruitTopping
HerbSpiceTopping
NutTopping
SauceTopping
VegetableTopping
Asserted class hierarchy
Inferred class hierarchy
Class Annotations: NonVegetarianTopping
Annotations
Class Annotations
Class Usage
negation - complement
Class Description: NonVegetarianTopping
Equivalent classes
PizzaTopping
and not VegetarianTopping
Superclasses
PizzaTopping
Inherited anonymous classes
Instances
Disjoint classes

e.g.(4pizza) Margherita

Translate ☺

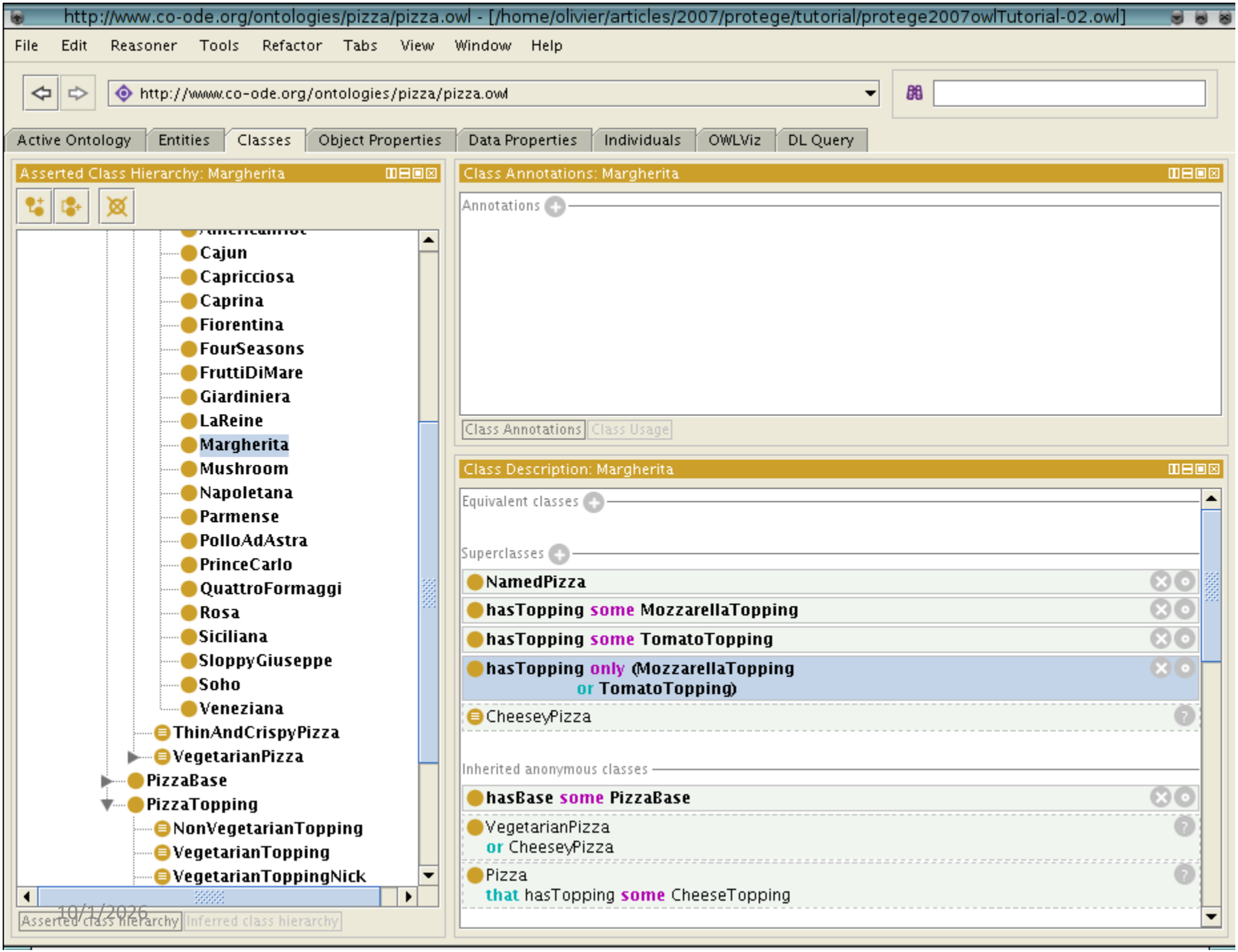
- Margherita pizzas **only** have Tomato and Mozzarella for topping

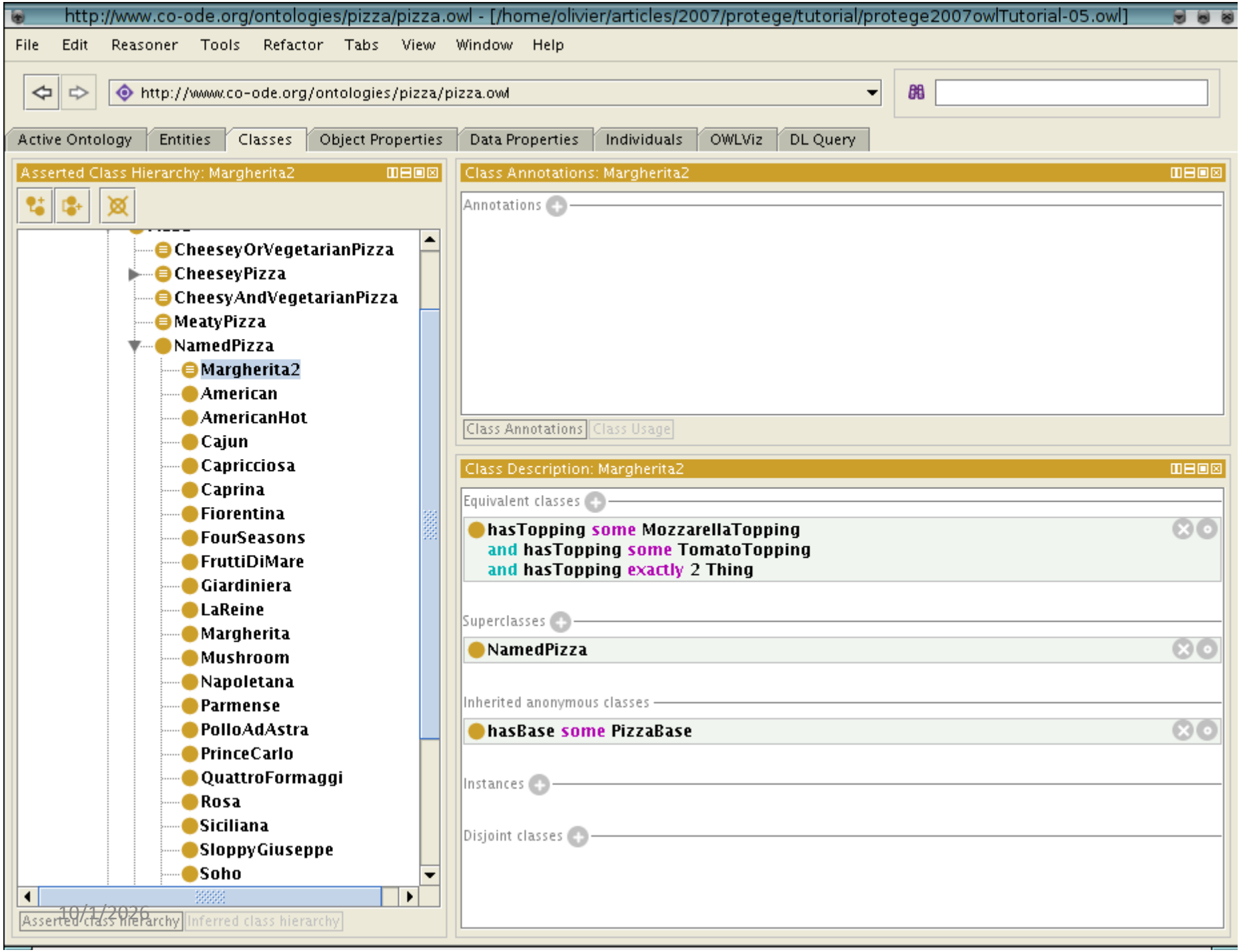
e.g.(4pizza) Margherita

Translate ☺

- Margherita pizzas have Tomato and Mozzarella for topping, and **only** these

- MargheritaPizza $\subset$ ($\exists$ *hasTopping* Mozzarella) $\cap$ ($\exists$ *hasTopping* Tomato) $\cap$ ($\forall$ *hasTopping* (Mozzarella U Tomato))


http://www.co-ode.org/ontologies/pizza/pizza.owl - [/home/olivier/articles/2007/protege/tutorial/protege2007owlTutorial-02.owl]
File Edit Reasoner Tools Refactor Tabs View Window Help
http://www.co-ode.org/ontologies/pizza/pizza.owl
Active Ontology Entities Classes Object Properties Data Properties Individuals OWLViz DL Query
Asserted Class Hierarchy: Margherita
Cajun
Capricciosa
Caprina
Fiorentina
FourSeasons
FruttiDiMare
Giardiniera
LaReine
Margherita
Mushroom
Napoletana
Parmense
PolloAdAstra
PrinceCarlo
QuattroFormaggi
Rosa
Siciliana
SloppyGiuseppe
Soho
Veneziana
ThinAndCrispyPizza
VegetarianPizza
PizzaBase
PizzaTopping
NonVegetarianTopping
VegetarianTopping
VegetarianToppingNick
Asserted class hierarchy
Inferred class hierarchy
Class Annotations: Margherita
Annotations
Class Annotations
Class Usage
Class Description: Margherita
Equivalent classes
Superclasses
NamedPizza
hasTopping some MozzarellaTopping
hasTopping some TomatoTopping
hasTopping only (MozzarellaTopping or TomatoTopping)
CheeseyPizza
Inherited anonymous classes
hasBase some PizzaBase
VegetarianPizza or CheeseyPizza
Pizza that hasTopping some CheeseTopping
10/1/2026


http://www.co-ode.org/ontologies/pizza/pizza.owl - [/home/olivier/articles/2007/protege/tutorial/protege2007owlTutorial-05.owl]
File Edit Reasoner Tools Refactor Tabs View Window Help
http://www.co-ode.org/ontologies/pizza/pizza.owl
Active Ontology Entities Classes Object Properties Data Properties Individuals OWLViz DL Query
Asserted Class Hierarchy: Margherita2
CheeseyOrVegetarianPizza
CheeseyPizza
CheesyAndVegetarianPizza
MeatyPizza
NamedPizza
Margherita2
American
AmericanHot
Cajun
Capricciosa
Caprina
Fiorentina
FourSeasons
FruttiDiMare
Giardiniera
LaReine
Margherita
Mushroom
Napoletana
Parmense
PolloAdAstra
PrinceCarlo
QuattroFormaggi
Rosa
Siciliana
SloppyGiuseppe
Soho
Asserted class hierarchy
Inferred class hierarchy
Class Annotations: Margherita2
Annotations
Class Annotations
Class Usage
Class Description: Margherita2
Equivalent classes
hasTopping some MozzarellaTopping
and hasTopping some TomatoTopping
and hasTopping exactly 2 Thing
Superclasses
NamedPizza
Inherited anonymous classes
hasBase some PizzaBase
Instances
Disjoint classes

e.g.(Riddles)

https://www.brainzilla.com/logic/zebra/einsteins-riddle/

<u>small e.g.</u>

? Japaneze, horse, blue

There are three houses.
In the green house there is a snake.
The American owns the dog.
The Englishman lives in the red house.

Now, who lives in the blue house? What animal owns the Japanese?
In the interest of clarity, it must be added that each of the three houses is painted a different colour, and their inhabitants are of different nationality and own different pets.
There is no ordering of the house.

e.g.(Riddles)

small e.g.

? Japaneze, horse, blue

There are three houses.
In the green house there is a snake.
The American owns the dog.
The Englishman lives in the red house.

Prolog solution: (https://swish.swi-prolog.org/)

```
house(englishman, _, red).
house(american, dog, _).
house(_,snake,green).
```

Queries:

```
house(X,_,blue).                X = american
house(japanese,X,_).            X = snake
```

But: Missing explicit specification that there are three houses

```
house(X,_,_).          =>       X = englishman
                                X = american
                                true
```

•Este un **nucleu al Web-ului de Date Deschise**, citat frecvent în cercetare și folosit în aplicații AI.
•Permite **interogări complexe** pe datele Wikipedia, imposibile direct pe site.
•Contribuie la interoperabilitatea datelor la scară globală.
Dacă vrei, pot să-ți arăt cum să interoghezi DBpedia cu SPARQL sau cum se compară mai detaliat cu Wikidata.